

One- way ANalysis Of Variance

Chapter 3

Key Design words

- **Response variable, Constants, Random noise**
- **Factors:** Qualitative (Nominal or Ordinal) or Quantitative
- **Operating window**
- **Levels** (for each Factor)
- **Treatment** (combination of levels)
- **Replicates** (usually constant for each treatment)
- **Run**
- **Total number of runs (Size of the experiment)**

Process Capability Ratio

Process

Y Quality Characteristic (Response of interest)

TI Tolerance Interval = USL - LSL

Y Deterministic or Stochastic?

Inside or Outside the TI ?

Process Capability Ratio

Tabella 7.2 Valori dell'indice di capacità (C_p) e corrispondenti tassi di difettosità per il caso di distribuzione normale (ppm).

C_p	Difettosi nel processo (ppm)	
	Specifiche unilaterali	Specifiche bilaterali
0.25	226628	453255
0.50	66807	133614
0.60	35931	71861
0.70	17865	35729
0.80	8198	16395
0.90	3467	6934
1.00	1350	2700
1.10	484	967
1.20	159	318
1.30	48	96
1.40	14	27
1.50	4	7
1.60	1	2
1.70	0.17	0.34
1.80	0.03	0.06
2.00	0.0009	0.0018

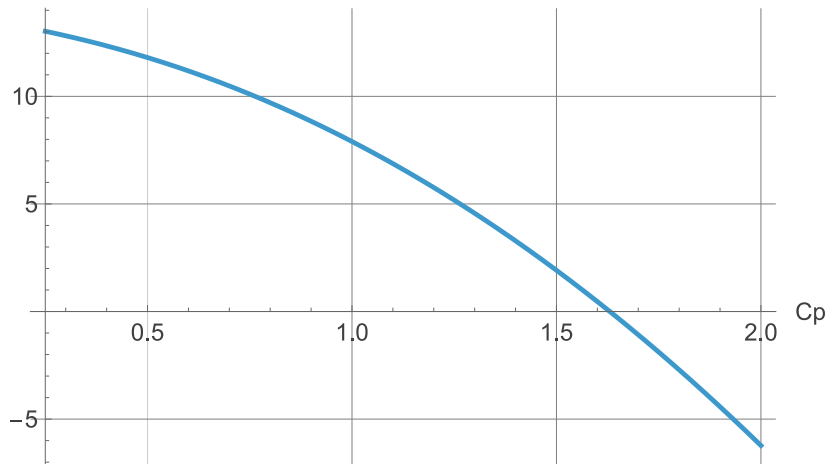
$$PCR \equiv C_p = \frac{USL - LSL}{6\sigma}$$

find the formula

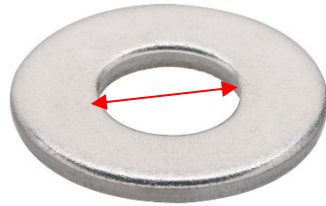
7.3 Valori minimi raccomandati per l'indice di capacità di un processo.

	Specifiche bilaterali	Specifiche unilaterali
Processi già esistenti	1.33	1.25
Processi nuovi	1.50	1.45
Parametri di sicurezza, di resistenza o critici di processi esistenti	1.50	1.45
Parametri di sicurezza, di resistenza o critici di processi nuovi	1.67	1.60

Ln(defect(ppm))



washer



Knurled shaft



Example 1

- Context / hypotheses

- Product: C30 Steel pin;
- Quality characteristic: Nominal diameter $D \pm d$ ($100 \pm 0.03\text{mm}$)
- Mass production: (2 automatic lathes)
- Cycle time: 3s/piece
- The manufactured diameter is distributed as a Gaussian distribution
- PCR=1.5 (7 ppm)

- Problem definition: Objective+Factors

- We want to check if the two machines are equally tuned (**why?**
could I consider different objectives?)

WHY?

- Let us suppose that the pin batch comes out **equally mixed** from the two machines (50% + 50%)
- Let us suppose that the two machines produce pins with different variances and averages
- You can easily prove that if:

$$Y_1 \rightarrow E(Y_1) = \mu_1 \quad V(Y_1) = \sigma_1^2$$

$$Y_2 \rightarrow E(Y_2) = \mu_2 \quad V(Y_2) = \sigma_2^2$$

$$E(Y) = \frac{1}{2} (E(Y_1) + E(Y_2)) = \frac{\mu_1 + \mu_2}{2}$$

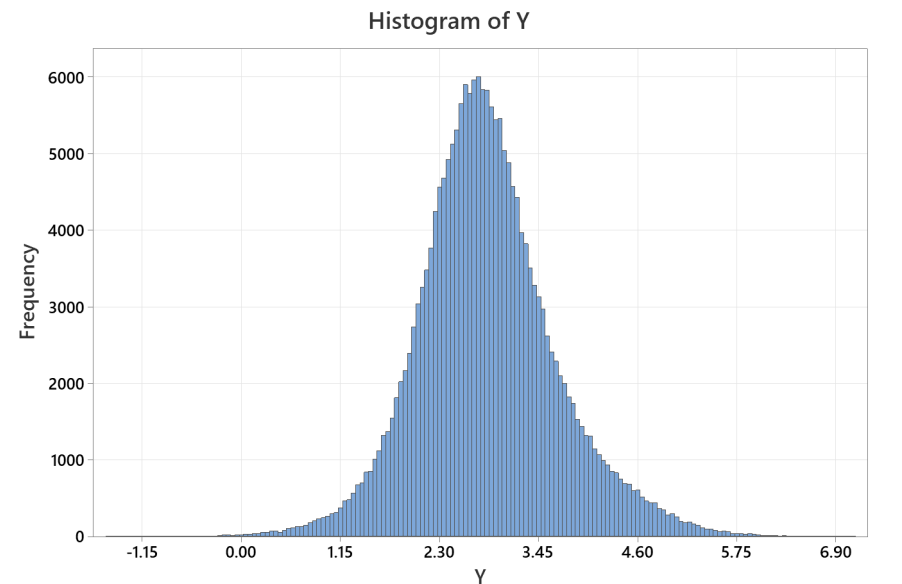
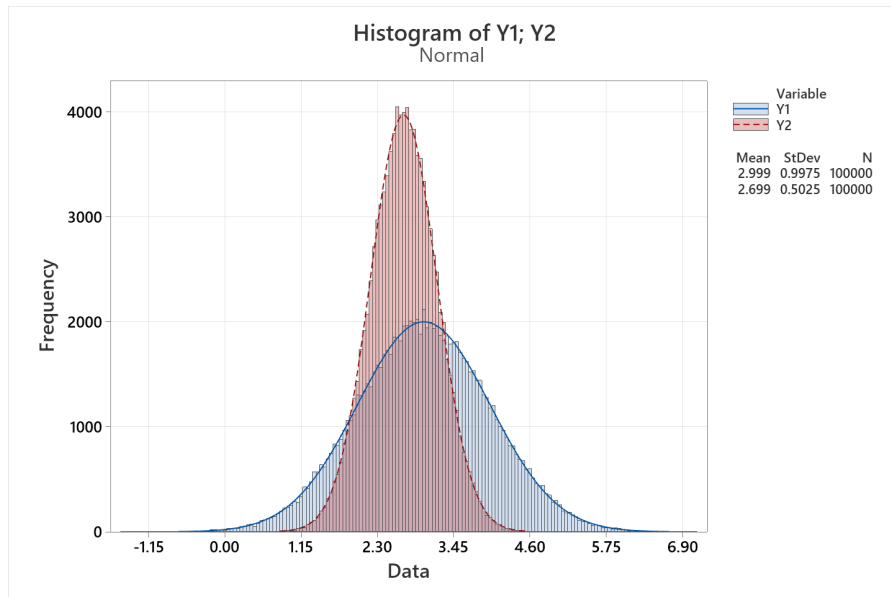
$$V(Y) = \frac{\sigma_1^2 + \sigma_2^2}{2} + \frac{\delta_\mu^2}{4}$$

$$\delta_\mu = \mu_1 - \mu_2$$

Simulation Experiment (Monte Carlo Method) with Gaussian Distributions (the Gaussian distribution is not strictly necessary)

$$Y_1 \sim NID(3.0; 1^2) \quad Y_2 \sim NID(2.7; 0.5^2) \quad n_1 = n_2 = 100000$$

$$E(Y) = \frac{\mu_1 + \mu_2}{2} = \frac{3 + 2.7}{2} = 2.85 \quad V(Y) = \frac{\sigma_1^2 + \sigma_2^2}{2} + \frac{\delta_\mu^2}{4} = \frac{1^2 + 0.5^2}{2} + \frac{(3.0 - 2.7)^2}{4} = 0.6475$$



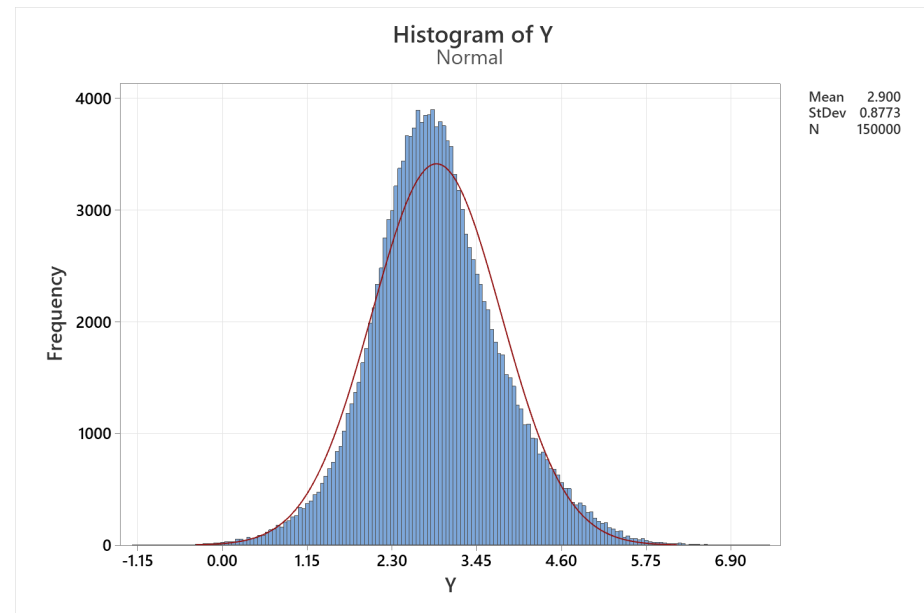
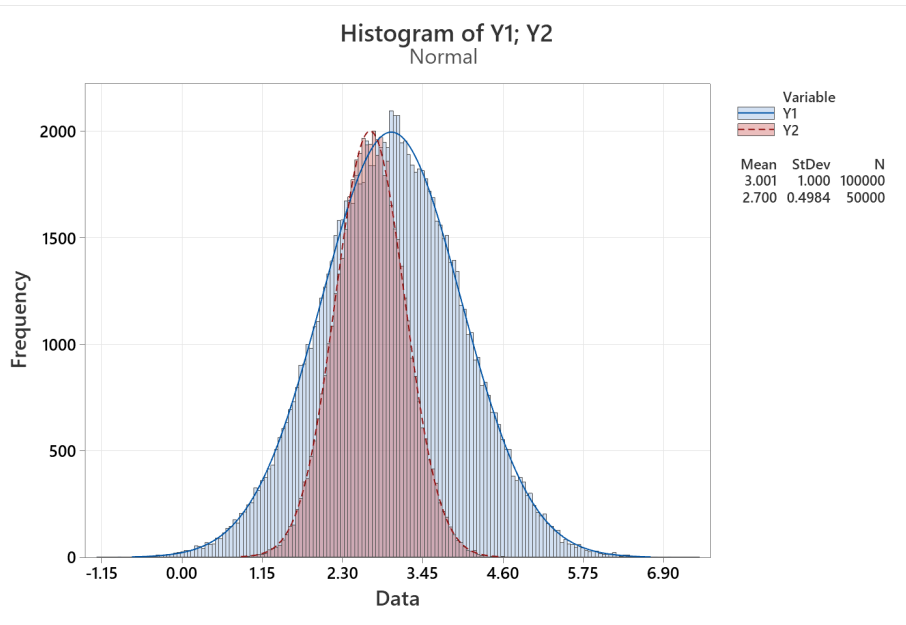
Statistics

Variable	Mean	Variance
Y	2.8493	0.6463

Imbalanced case

$$Y_1 \sim NID(3.0, 1^2) \quad Y_2 \sim NID(2.7, 0.5^2) \quad n_1 = 100000 \quad n_2 = 50000$$

Theoretical formulas? **make them yourself**



Statistics

Variable

Y

Mean Variance

2.9005 0.7696

- Experiment

- **Measuring:** measure two orthogonal diameters using a caliper (**is the measurement process adequate?**)
- **Experimenting:** Take a random sample from the daily production: $N=n_1+n_2$ pins (**how do I define these values?**)

- Data Analysis

- Use **EDA**; check the distribution; **Two -Sample t Test** to draw some conclusion

- Context / hypotheses
 - Product: C30 Steel pin;
 - Quality characteristic: nominal diameter $D \pm d$ ($100 \pm 0,03\text{mm}$)
 - Mass production (**4** automatic lathes)
 - Cycle time: 3s/piece
 - The manufactured diameter is distributed as a Gaussian distribution
 - PCR=1.5 (7 ppm)
- Problem definition: Objective+Factors
 - We want to check if the **4** machines are equally tuned (**why? could I consider different objectives?**)
- Experiment
 - **Measuring**: measure two orthogonal diameters using a caliper (**is the measurement process adequate?**)
 - **Experimenting**: Take 4 random samples from the daily production: $N = n_A + n_B + n_C + n_D$ pins (**how do I define these values?**)
- Data Analysis
 - Use **EDA**; check the distribution; **may I use the Two-Sample t-Test for each couple of samples?**

Test errors (One shot test)

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2$$

- We set a **statistic S** and a **critical test region w_0** such that

if $S \in w_0 \rightarrow \text{Reject } H_0$

$$\alpha = \Pr(\text{Reject } H_0 | H_0 \text{ is True})$$

$$\beta = \Pr(\text{Fail to reject } H_0 | H_1 \text{ is True})$$

STATE OF THE NATURE

ACTION

	H_0 true	H_1 true
Reject H_0	α	Test Power $1 - \beta$
Fail to reject H_0	$1 - \alpha$	β

- Let us suppose we install an alarm at home
- Let us tune the alarm in order to have a false alarm at 5% rate (**alpha error**).
- In other words, on the average, we will wake up 5 nights out of 100 without a burglar breaking into your home.
- But we are terribly afraid of burglars and we install 100 alarm systems, each tuned at 5% rate.
- The question is: **Will you sleep?**
- The same applies when you have several tests to make: **you must turn to the Alpha familywise concept**

Let us consider the **Family of tests** to check if the **statistic S_i** belongs or not to the **critical test region w_{0i}**

If the null hypothesis H_{0i} is true, we define the following events

$$A_i : S_i \in w_{0i}$$

$$\Pr(A_i | H_{0i}) = \alpha_i$$

$$\bar{A}_i : S_i \notin w_{0i}$$

$$\Pr(\bar{A}_i | H_{0i}) = 1 - \alpha_i$$

Familywise error of first type α_{FAM} is the probability of **rejecting at least** one hypothesis if **all are true**

$$\alpha_{FAM} = \Pr(A_1 \cup A_2 \cup \dots \cup A_i \cup \dots | H_{01}, H_{02}, \dots, H_{0i}, \dots)$$

Assume that $\alpha_i = \alpha \quad \forall i$

Completely dependent events

$$\alpha_{FAM} = \Pr(A_1 \cup A_2 \dots \cup A_p \mid H_{01}, H_{02} \dots H_{0p}) = \alpha$$

Independent events

$$1 - \alpha_{FAM} = (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_p)$$

$$SIDAK \quad \alpha_{FAM} = 1 - \prod_{i=1, \dots, p} (1 - \alpha_i)$$

$$\alpha_i = \alpha \quad \forall i = 1, \dots, p$$

$$\alpha_{FAM} = 1 - (1 - \alpha)^p \quad \longrightarrow \quad \alpha = 1 - (1 - \alpha_{FAM})^{1/p}$$

Unknown dependency structure

Bonferroni inequality

(for unknown dependency)

PROF. BONFERRONI

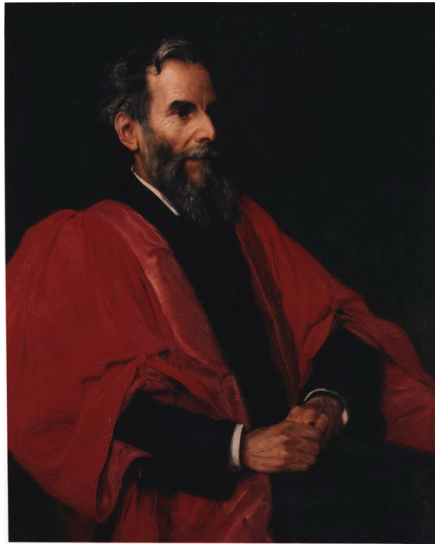


*Siccome si è raccomandato
perchè non lo sfottessimo
non possiamo far altro che obbedire.*



Carlo Emilio Bonferroni
1892-1960

Unknown dependency

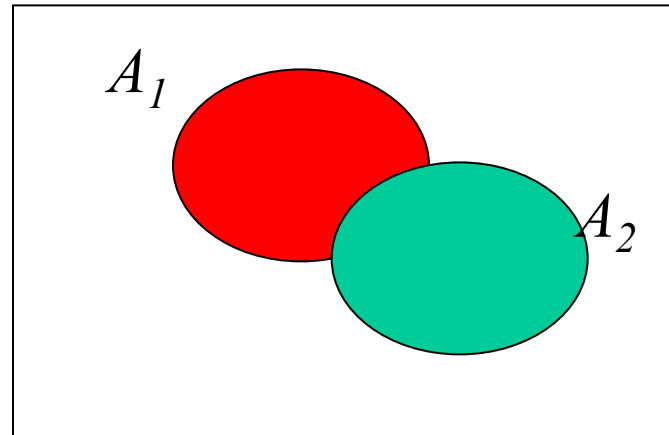


John Venn
1834-1923

$$A_1 : S_1 \in w_{01} \quad \Pr(A_1 | H_{01}) = \alpha_1$$

$$A_2 : S_2 \in w_{02} \quad \Pr(A_2 | H_{02}) = \alpha_2$$

$$\alpha_{FAM} = \Pr(A_1 \cup A_2 | H_{01}, H_{02})$$



$$A_1 \cup A_2$$

$$\begin{aligned} \alpha_{FAM} &= \Pr(A_1 \cup A_2 | H_{01}, H_{02}) = \Pr(A_1 | H_{01}) + \Pr(A_2 | H_{02}) - \Pr(A_1 \cap A_2 | H_{01}, H_{02}) \\ &\leq \Pr(A_1 | H_{01}) + \Pr(A_2 | H_{02}) \end{aligned}$$

Bonferroni Inequality

$$\alpha_{FAM} \leq \sum_{i=1, \dots, p} \alpha_i$$

Unknown dependency

Bonferroni Inequality

$$\alpha_{FAM} \leq \sum_{i=1, \dots, p} \alpha_i \quad \Rightarrow \quad \alpha_{FAM} \leq p\alpha$$

Could we use the Bonferroni Inequality to limit the α_{FAM} ?

a) Set α_{FAM}^*

b) For each test, use $\alpha = \frac{\alpha_{FAM}^*}{p}$

Bonferroni inequality states that $\alpha_{FAM} \leq \alpha_{FAM}^*$

Single factor, many levels

- We can use the **Two-Sample t -Test** (Student test) for each pair of levels using the **Bonferroni approach**.
- Unfortunately the test **Power** decreases as the number of levels increases.
- **Analysis of Variance (ANOVA)** is the appropriate tool
- **ANOVA** was developed by Sir Ronald **FISHER** in the 20-30s for Agricultural experiments.
- Now **ANOVA** is applied everywhere

Example

- We are interested in the Cotton weight percentage of a new synthetic fiber for shirts.
- The Response variable (**Quality Characteristic**) is the **Tensile strenght** (**higher is better**)
- We are interested in finding if the **Cotton weight percentage (Factor)** influences the **Tensile strenght (Response variable)**
- The weight cotton content should be between about 10% and 40% (**Factor Range or Operating window definition**)
- We suspect that increasing the weight cotton percentage, the tensile strenght will increase at least initially (**previous information**)
- Questions:
 - Does the tensile strenght **change** as much as the cotton weight percentage change?
 - Does an **optimal level exist**?

- **Response Variable:** Tensile strenght
- The experimenter chooses the **Factor Range** [5-35]%
- He/she defines **5 levels (treatment):** [15, 20, 25, 30, 35] %
- He/she decides to replicate **5** times each level (treatment). The Experiment size is **N=25**
- The **runs** are performed in a **random order**

Table 3-1 Data (in lb/in²) from the Tensile Strength Experiment

Cotton Weight Percentage	Observations					Total	Average
	1	2	3	4	5		
15	7	7	15	11	9	49	9.8
20	12	17	12	18	18	77	15.4
25	14	18	18	19	19	88	17.6
30	19	25	22	19	23	108	21.6
35	7	10	11	15	11	54	10.8
						376	15.04

Table 3-2 Typical Data for a Single-Factor Experiment

Treatment (level)	Observations				Totals	Averages
1	y_{11}	y_{12}	$\dots$	y_{1n}	$y_{1.}$	$\bar{y}_{1.}$
2	y_{21}	y_{22}	$\dots$	y_{2n}	$y_{2.}$	$\bar{y}_{2.}$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$
a	y_{a1}	y_{a2}	$\dots$	y_{an}	$y_{a.}$	$\bar{y}_{a.}$
					$y_{..}$	$\bar{y}_{..}$

- In general we will have **a Factor levels**, (**a Treatments**), and **n Replicates** (The math can tackle the case in which **n** changes at each level)
- The **Factor** is a variable
- The **runs** should be performed in **Random order: CRD Complete Randomized Design**
- **N = an** Total number of **runs (Size of the experiment)**
- We have two kinds of model: **Fixed effects** or **Random effects**

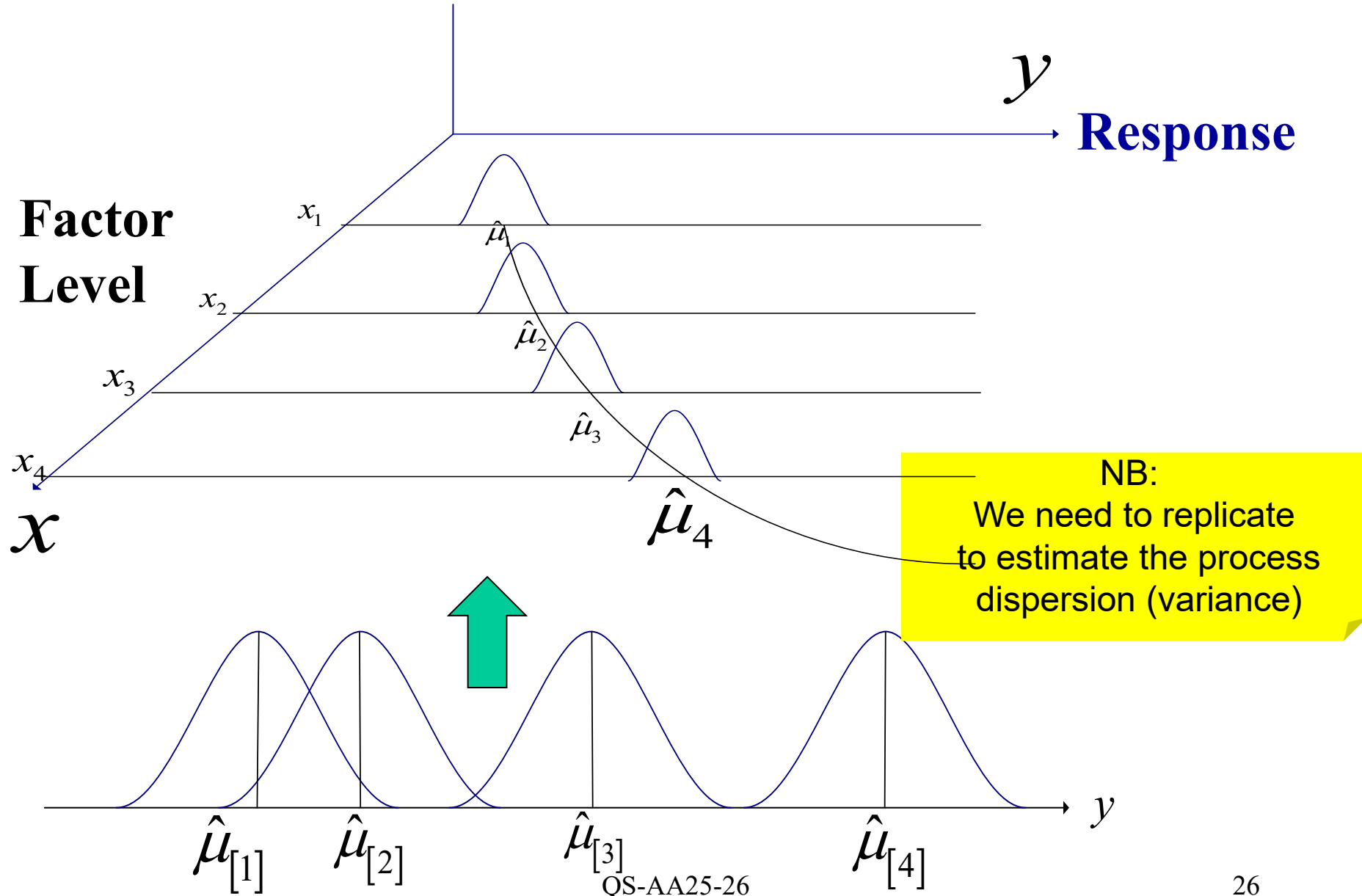
Usual Data analysis goal:

- Does the Factor influence the Response?.**
- Could we define one or more optimal Factor levels?**

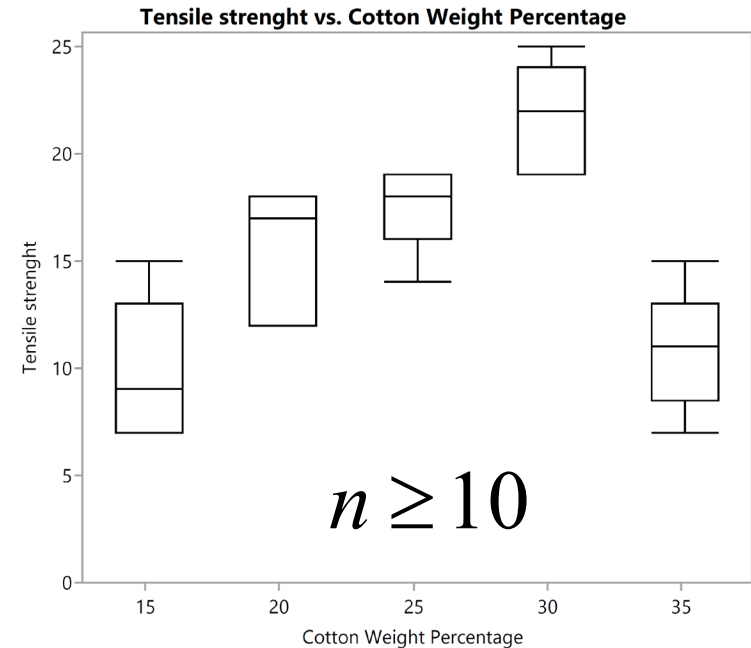
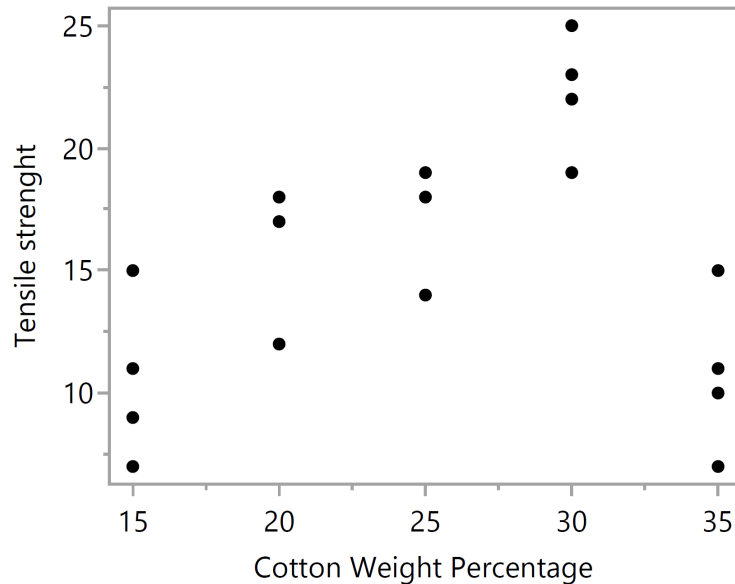
Other examples

- What's the **colour** cheering up people in Hospital;
- Study the **success of suicide** by jumping from different building windows
- To design the experiment, you have to define:
 - Response Variable and how to measure it
 - Factor and its levels
 - The number of replications

Regression vs ANOVA Fixed effects model



Back to the example: EDA

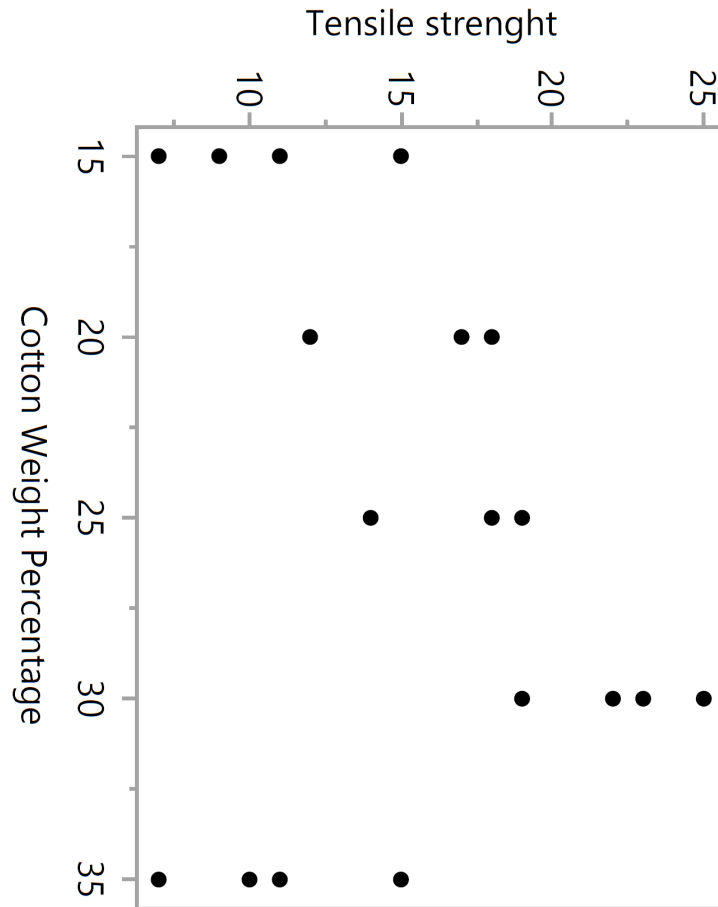


Goals:

- **Outliers detection**
- **Hypothesis checking**
- **Goal:** Statistical Different means $\rightarrow$ Factor significance

My favorite phrase: **You must torture your data set, in the end it will confess!**

One Way ANOVA



- In “**Analysis of Variance**” –**ANOVA**- we split the **total variability** of the **observed data** in **components (sources)** associated with the **model parameters**
- Possible models (**Fixed effects**) are:
 - Model of Means $y_{ij} = \mu_i + \varepsilon_{ij}$
 - Model of Effects $y_{ij} = \mu + \tau_i + \varepsilon_{ij}$
 - Regression Model

$$y_{ij} = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_a x_a + \varepsilon_{ij}$$

with $x_i \in \{0, 1\}$

Model Hypotheses

- The Hypotheses are on the ε_{ij} (usually called **Model errors** but they are the **Random component of the Model**)
- They should be:
 - Normally distributed
 - Null Expected value
 - Constant variance (**between** and **within** the factor levels)
 - Indipendency (**between** and **within** the levels)
- Summing up: $\varepsilon_{ij} \sim NID(0, \sigma^2)$

Model parameter estimate

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij} \longrightarrow y_{ij} = \hat{\mu} + \hat{\tau}_i + e_{ij}$$

$$y_{ij} = \mu_i + \varepsilon_{ij} \longrightarrow y_{ij} = \hat{\mu}_i + e_{ij}$$

Model parameter estimate

- **Minimum of OLS**

(OLS: Ordinary Least Squares)

$$Q = \sum_{i,j} \varepsilon_{ij}^2 = \sum_{i,j} (y_{ij} - \mu - \tau_i)^2$$

Normal Equations

$$\frac{\partial Q}{\partial \mu} = 0 \quad -2 \sum_{i,j} (y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0 \quad \longrightarrow \quad na\hat{\mu} + n\hat{\tau}_1 + \dots + n\hat{\tau}_a = y_{..}$$

$$\frac{\partial Q}{\partial \tau_i} = 0 \quad -2 \sum_j (y_{ij} - \hat{\mu} - \hat{\tau}_i) = 0 \quad \longrightarrow \quad \begin{array}{l} n\hat{\mu} + n\hat{\tau}_1 = y_{1.} \\ \dots\dots\dots \\ n\hat{\mu} + n\hat{\tau}_i = y_{i.} \end{array}$$

Residuals $\longrightarrow e_{ij} = y_{ij} - \hat{\mu} - \hat{\tau}_i$

Normal Equations

$$y_{..} = an\hat{\mu} + n\hat{\tau}_1 + n\hat{\tau}_2 + \dots + n\hat{\tau}_a$$

$$y_{1.} = n\hat{\mu} + n\hat{\tau}_1$$

$$y_{2.} = n\hat{\mu} + n\hat{\tau}_2$$

$$y_{a.} = n\hat{\mu} + n\hat{\tau}_a$$

Unfortunately these Equations are **linearly dependent**.

To solve them we have to add **constraints**

In this case just **one constraint**: i.e.

$$\sum_{i=1,a} \hat{\tau}_i = \hat{\tau}_1 + \hat{\tau}_2 + \dots + \hat{\tau}_a = 0$$

Note: n is constant, but Math can tackle different values of n_i

The Normal Equation **solutions** are:

$$\hat{\mu} = \frac{y_{..}}{an} = \bar{y}_{..}$$
$$\hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..}$$

Note: whatever the constraint, we have:

$$\hat{\mu}_i = \hat{\mu} + \hat{\tau}_i = \bar{y}_{i.}$$

Please try with other constraints

Rules to write down the Normal Equations

- **Rule #1**
 - One Normal Equation for each model parameter
- **Rule # 2**
 - The **left-hand side** is just the sum of all observations containing the particular parameter associated with that particular normal equation.
- **Rule #3**
 - The **right-hand side** is the sum of all model parameters where each parameter is multiplied by the number of times it appears in the total on the left-hand side.

Rule 1

$$\begin{array}{l} \hat{\mu} \mapsto y_{..} = an\hat{\mu} + n\hat{\tau}_1 + n\hat{\tau}_2 + \dots\dots\dots + n\hat{\tau}_a \\ \hat{\tau}_1 \mapsto y_{1.} = n\hat{\mu} + n\hat{\tau}_1 \\ \hat{\tau}_2 \mapsto y_{2.} = n\hat{\mu} \qquad \qquad \qquad + n\hat{\tau}_2 \\ \dots\dots\dots \\ \hat{\tau}_a \mapsto y_{a.} = n\hat{\mu} \qquad \qquad \qquad + n\hat{\tau}_a \end{array}$$

Rule 2

Rule 3

Model hypothesis checking

- The **residuals** are used to check the model hypotheses

$$e_{ij} = y_{ij} - \hat{\mu} - \hat{\tau}_i \quad \text{or (by substitution)} \quad e_{ij} = y_{ij} - \hat{y}_{ij}$$

- What checks?** $\text{FIT} \rightarrow \hat{y}_{ij} = \hat{\mu}_i = \hat{\mu} + \hat{\tau}_i = \bar{y}_i.$
 - Residual graphs (Usually SRES vs FITS)
 - Extreme residuals
 - Normality assumption
 - Constant variance
 - Statistical Indipendency (**Run Order Indipendency** only) Consider that we do not have N independent residuals as we have a constraints on them. This implies that $df_E = N - a$

ANOVA (Analysis of Variance)

- **ANOVA** splits the **total variability** of the **data** in components (sources) associated with each model parameter
- We start from $y_{ij} = \hat{\mu} + \hat{\tau}_i + e_{ij}$

$$(y_{ij} - \hat{\mu}) = \hat{\tau}_i + e_{ij}$$

- **Squaring and summing** we have:

$$\sum_{i=1,a} \sum_{j=1,n} (y_{ij} - \hat{\mu})^2 = \sum_{i,j} \hat{\tau}_i^2 + \sum_{i,j} e_{ij}^2 + \sum_{i,j} 2\hat{\tau}_i e_{ij}$$

$$\sum_{i=1,a} \sum_{j=1,n} (y_{ij} - \hat{\mu})^2 = n \sum_i \hat{\tau}_i^2 + \sum_{i,j} e_{ij}^2$$

- **Note: it's possible to prove that mixed product terms are null**

$$\sum_{i=1,a} \sum_{j=1,n} (y_{ij} - \hat{\mu})^2 = n \sum_i \hat{\tau}_i^2 + \sum_{i,j} e_{ij}^2$$

$$\sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{..})^2 = n \sum_{i=1}^a (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^a \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

$$SS_{TOT} = SS_{FAC} + SS_E$$

- A big SS_{FAC} value implies, probably, level differences
- A small SS_{FAC} value implies, probably, no level difference
- The **statistical hypotheses** to test are:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_a \text{ or } H_0 : \tau_1 = \tau_2 = \dots = \tau_a = 0$$

$$H_1 : \text{at least a couple of means are different}$$

$$\sum_{i=1,a} \sum_{j=1,n} (y_{ij} - \hat{\mu})^2 = n \sum_i \hat{\tau}_i^2 + \sum_{i,j} e_{ij}^2$$

↓

$SS_{TOT(\text{corrected})}$

↓

SS_{FAC}

↓

SS_E
or BETWEEN or WITHIN

$$(N - 1) = (a - 1) + (N - a)$$

$$MS_{FAC} = \frac{SS_{FAC}}{df_{FAC}}$$

$$df_{FAC} = a - 1$$

$$MS_E = \frac{SS_E}{df_E}$$

$$df_E = (N - a)$$

- **Variance estimate** for each level $\hat{\sigma}_i^2 = s_i^2 = \sum_j e_{ij}^2 / (n_i - 1)$
- If $n = \text{constant}$ $\hat{\sigma}_i^2 = s_i^2 = \sum_j e_{ij}^2 / (n - 1)$

$$SS_E = \sum_{i=1,a} \left(\sum_{j=1,n} e_{ij}^2 \right) = \sum_{i=1,a} (n-1) s_i^2 = (n-1)a \sum_{i=1,a} \frac{s_i^2}{a} \rightarrow \sum_{i=1,a} \frac{s_i^2}{a} = \frac{SS_E}{(n-1)a} = \frac{SS_E}{N-a}$$

- If we assume that the **variance is constant** between the levels, we can estimate the **pooled variance**

$$s_{pooled}^2 = \sum_{i=1,a} \frac{s_i^2}{a} \quad s_{pooled}^2 = \frac{SS_E}{N-a} = \frac{SS_E}{df_E} = MS_E$$

Similarly

$$MS_{FAC} = \frac{SS_{FAC}}{a-1} = \frac{SS_{FAC}}{df_{FAC}}$$

ANOVA Table (One Way)

Source	SS Sum of squares	df degrees of freedom	MS mean squares	F_0
Factor (Between)	SS_{FAC}	$df_{FAC} = a - 1$	MS_{FAC}	$F_0 = \frac{MS_{FAC}}{MS_E}$
Error (Within)	SS_E	$df_E = a(n - 1) = N - a$	MS_E	
Total (corrected)	SS_{TOT}	$an - 1 = N - 1$		

$$SS_{TOT} = \sum_{i,j} y_{ij}^2 - \frac{y_{..}^2}{an} \qquad SS_{FAC} = \sum_{i,j} \frac{y_{i.}^2}{n} - \frac{y_{..}^2}{an}$$

$$SS_E = SS_{TOT} - SS_{FAC}$$

You can prove (exercise in ClassWork) that:

$$E(MS_{FAC}) = \sigma^2 + \frac{n}{a-1} \sum_{i=1,a} \tau_i^2$$

$$E(MS_E) = \sigma^2$$

$$H_0 : \tau_i = 0 \quad \forall i = 1, a$$



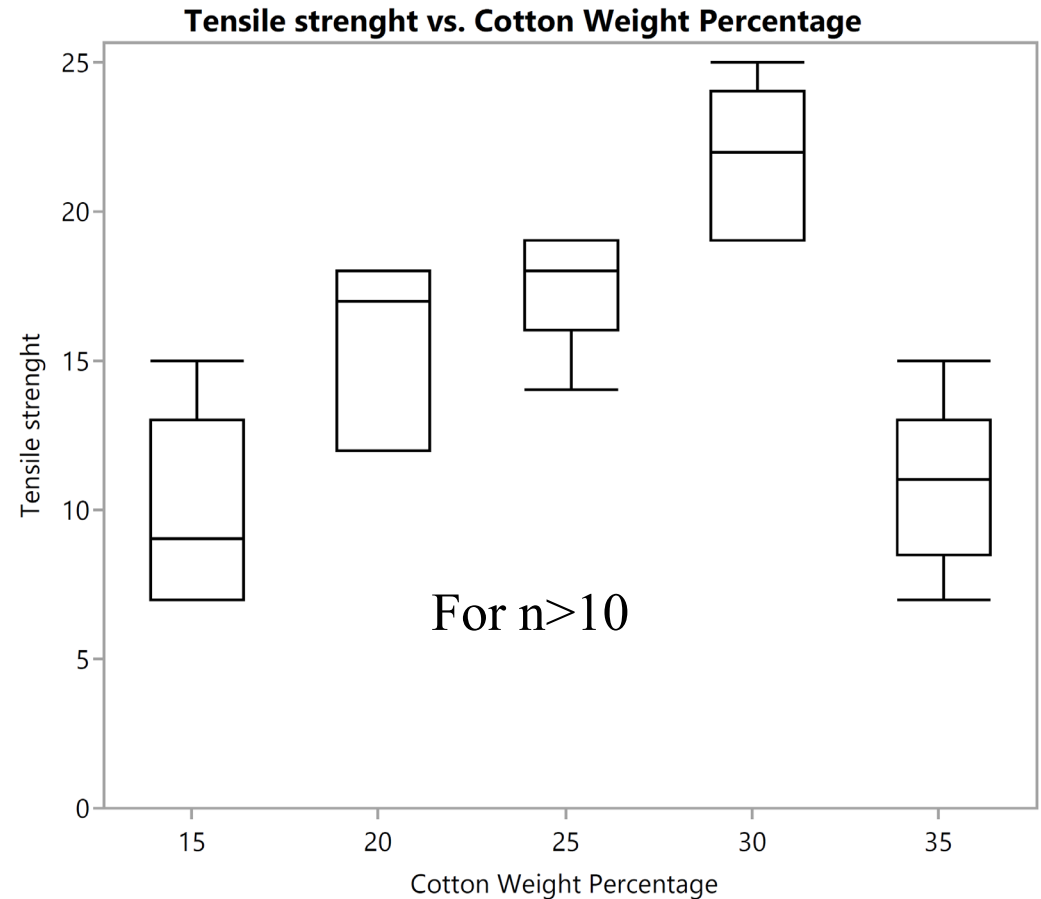
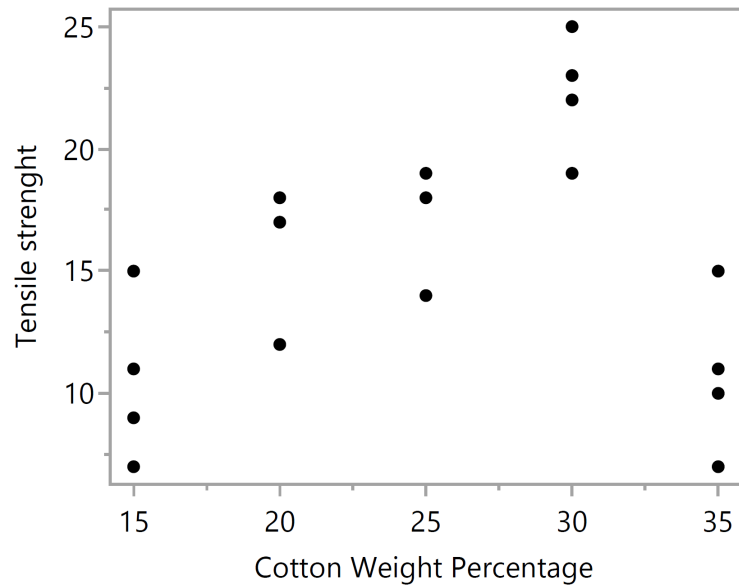
$$F_0 = \frac{MS_{FAC}}{MS_E} \sim F(a-1, df_E)$$

If $F_0 > F_\alpha(a-1, df_E)$
Reject H_0

Note the direction

or if $p_{value} \leq 5\%$ (or 1%)
Reject H_0

Back to the example: Data plot



Source	<i>df</i>	SS	MS	F_0	p
cotton %	4	475.76	118.94	14.76	0.000
Error	20	161.20	8.06		
Total	24	636.96			
Pooled StDev = 2.839					

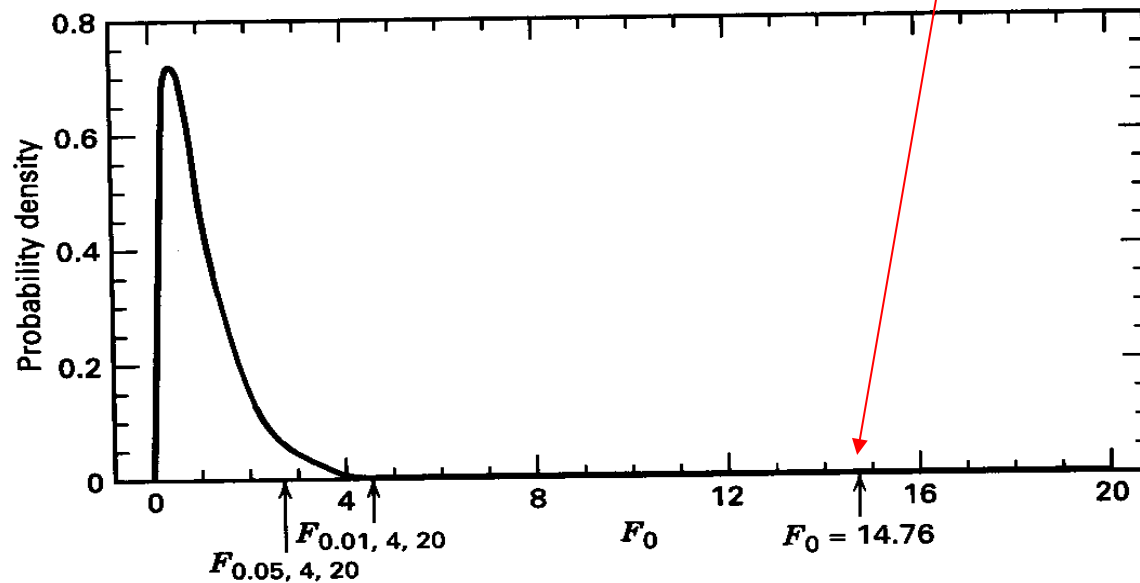


Figure 3-3 The reference distribution ($F_{4,20}$) for the test statistic F_0 in Example 3-1.

Testing the model hypotheses

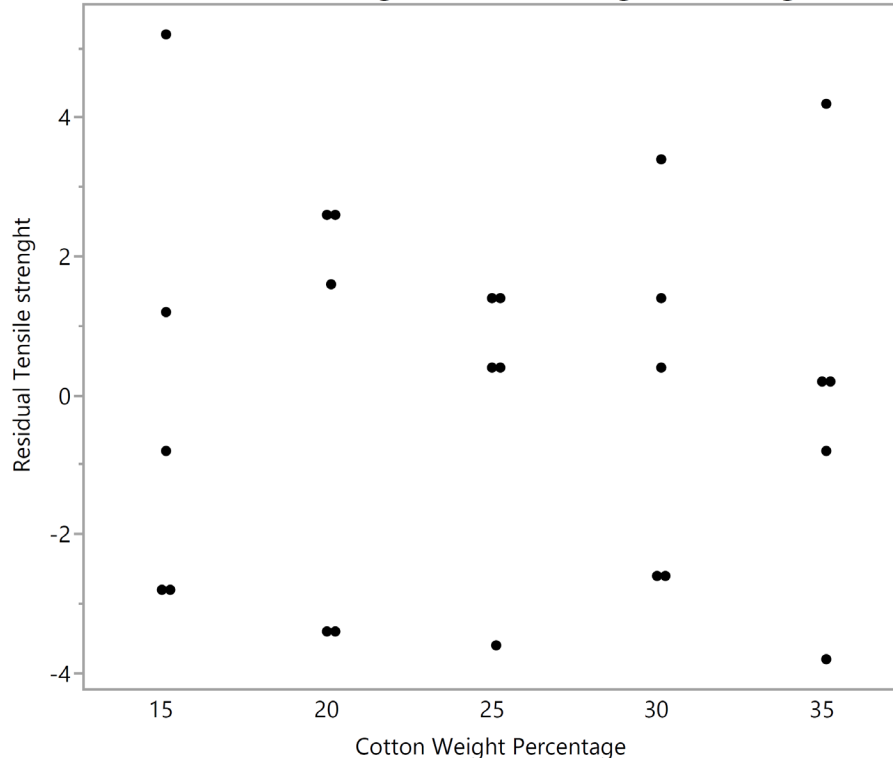
- Model **residuals** are used to check the model hypotheses

$$e_{ij} = y_{ij} - \hat{\mu}_i \quad \text{or} \quad e_{ij} = y_{ij} - \hat{y}_{ij}$$

- **What checks?**
 - Residual plot
 - Extreme residuals
 - Normality assumption
 - Constant variance
 - Statistical Indipendency (Run Order independency)

Residual graphs and Extreme values

Residual Tensile strenght vs. Cotton Weight Percentage

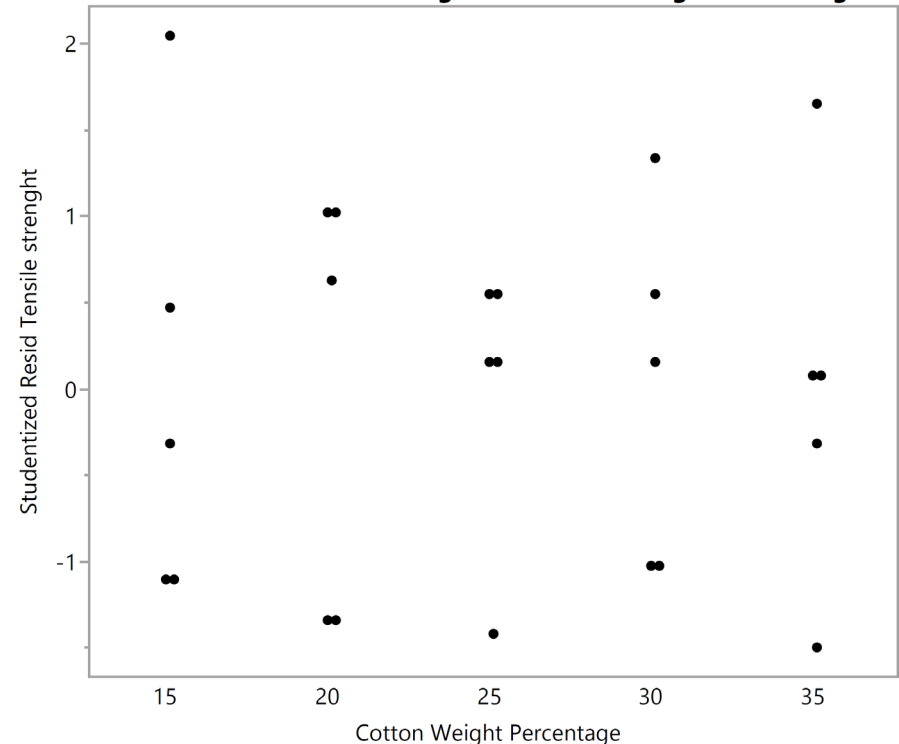


• Residual

$$SRES_{ij} = \frac{e_{ij}}{\sqrt{\hat{V}(e_{ij})}}$$

$$\hat{V}(e_{ij}) = \frac{n_i - 1}{n_i} MS_E \simeq MS_E = 8.06$$

Studentized Resid Tensile strenght vs. Cotton Weight Percentage

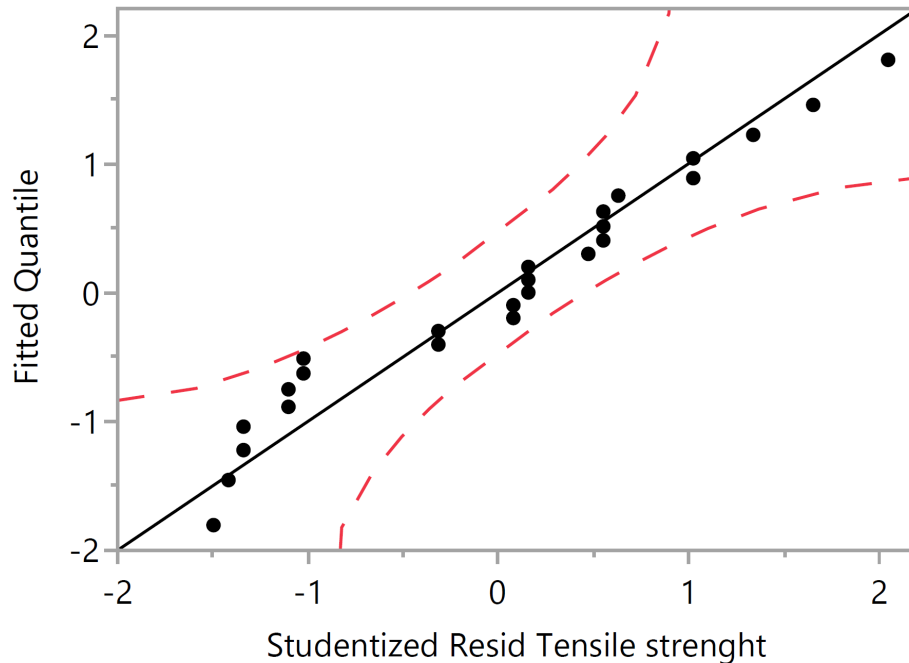


What residuals?

Studentized Residuals

Normality assumption checking

Normal Probability Plot of the Studentized Residuals



Goodness-of-Fit Test

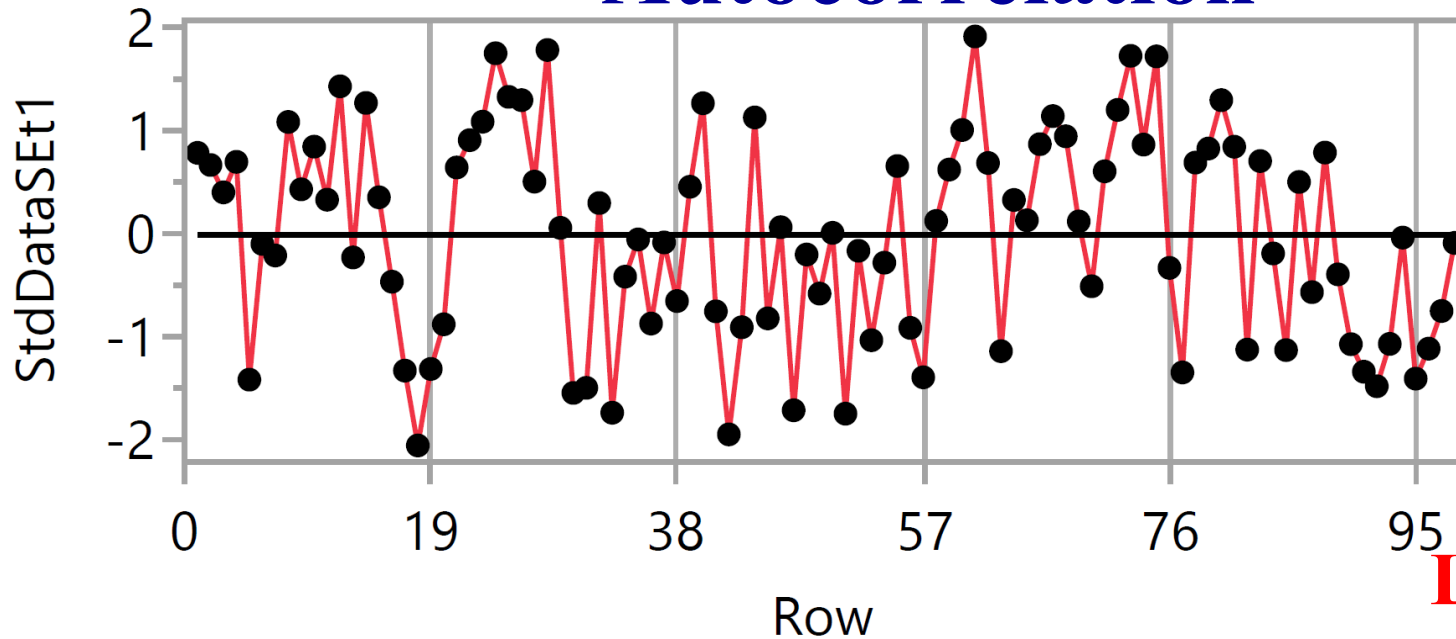
	W	Prob<W
Shapiro-Wilk	0.9438681	0.1818
	A ²	Simulated p-Value
Anderson-Darling	0.5185719	0.1680

Note: H_0 = The data is from the Normal distribution.
Small p-values reject H_0 .

Autocorrelation checking

- We should test the indipendency of the model errors
- The N residuals are **NOT** independent: they have $N-a$ *df*, but if a is smaller than N (i.e. 3 compared to 50) we can ignore this problem.
- We are interested in a particular dependency: the **temporal dependency (Run Order)** .
- To check this dependency we must order the residuals by the **Execution time** of each run (or **Run Order**) .
- To have a reasonable Test Power, we should have at least **50 observations**. In the following we will relax this constraint for the sake of simplicity.

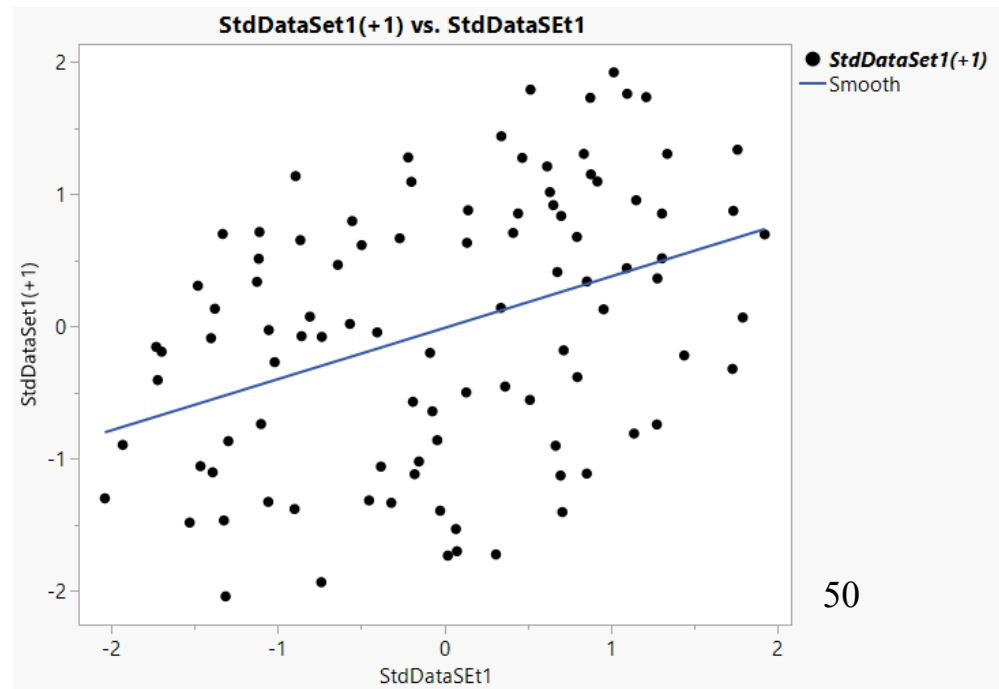
Autocorrelation



98 SRES

Lag 1 plot

The slope is a measure of
Linear Autocorrelation



50

We can change the **lag** (we call it **k**), and compute the slope again (for each **k**).

Theoretical Autocovariance function

$$\gamma_{t,k} = \text{Cov}(\varepsilon_t, \varepsilon_{t+k}) = E\left(\left(\varepsilon_t - E(\varepsilon_t)\right)\left(\varepsilon_{t+k} - E(\varepsilon_{t+k})\right)\right)$$

$$k = 0, \pm 1, \pm 2, \dots$$

Steady Process (mean and variance constant):

$$\gamma_{t,k} = \gamma_k = \gamma_{-k} \longrightarrow \gamma_k$$

$$E(\varepsilon_t) = E(\varepsilon) = 0$$

$$\gamma_k = E(\varepsilon_t \varepsilon_{t+k})$$

$$\text{For } k=0 \quad \gamma_0 = E\left[\left(\varepsilon - E(\varepsilon)\right)\right]^2 = E(\varepsilon^2) = V(\varepsilon) = \sigma^2$$

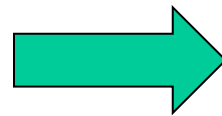
Usually we work with **dimensionless quantities**

AutoCorrelation Function (Pearson Correlation coefficient)

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\text{Cov}(\varepsilon_t, \varepsilon_{t+k})}{\text{Var}(\varepsilon_t)} \quad k = 0, \pm 1, \pm 2, \dots$$

$$-1 \leq \rho_k \leq 1$$

$$\gamma_k = \gamma_{-k} \Rightarrow \rho_k = \rho_{-k}$$



ACF plot for $k > 0$

General Proof that $|\rho| \leq 1$

Let's consider two random variables X and Y

$$V(X) = \sigma_X^2 \quad V(Y) = \sigma_Y^2 \quad \text{Cov}(X, Y) = \sigma_{XY}$$

- Let's consider the random variable Z (with a constant λ)

$$Z = X + \lambda Y$$

- The Variance of Z is $V(Z) = \lambda^2 \sigma_y^2 + 2\lambda \sigma_{xy} + \sigma_x^2$
- Where $\sigma_{xy} = \text{Cov}(X, Y)$
- By definition $V(Z)$ must be zero or positive $V(Z) \geq 0$
- This implies that the solution of the quadratic function in λ **can not** have distinct real roots

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \text{ with } a = \sigma_y^2 \quad b = 2\sigma_{xy} \quad c = \sigma_x^2$$

$$V(Z) \geq 0 \rightarrow 4\sigma_{xy}^2 - 4\sigma_x^2 \sigma_y^2 \leq 0 \rightarrow \sigma_{xy}^2 \leq \sigma_x^2 \sigma_y^2$$

- Consequently $\rho^2 = \frac{\sigma_{xy}^2}{\sigma_x^2 \sigma_y^2} \leq 1 \rightarrow |\rho| \leq 1$

Remarks:

- a) if ε_t and ε_{t+k} are independents, then they are not correlated ($\rho_k=0$)
- b) if ε_t e ε_{t+k} are correlated ($\rho_k \neq 0$ for some k), then **they are** dependents
- c) the considered correlation is a particular one: linear
- d) It exists a **Partial Autocorrelation Function PACF**

$$PACF(k) = \frac{Cov(x_t, x_{t-k} | x_{t-1}, x_{t-2}, \dots, x_{t-k+1})}{\sqrt{Var(x_t | x_{t-1}, x_{t-2}, \dots, x_{t-k+1}) Var(x_{t-k} | x_{t-1}, x_{t-2}, \dots, x_{t-k+1})}}$$

In this quantity you eliminate the influence of the terms between t and $t-k$

Practically we must use the **sampling Autocovariance and Autocorrelation** function :

$$\gamma_k = \text{Cov}(\varepsilon_t, \varepsilon_{t+k}) = E\left[(\varepsilon_t - E[\varepsilon_t])(\varepsilon_{t+k} - E[\varepsilon_{t+k}])\right] = E(\varepsilon_t \varepsilon_{t+k})$$

$$\hat{\gamma}_k = \frac{1}{N} \sum_{t=1}^{N-k} (e_t - \bar{e}_{\cdot})(e_{t+k} - \bar{e}_{\cdot}) = \frac{1}{N} \sum_{t=1}^{N-k} e_t e_{t+k}$$

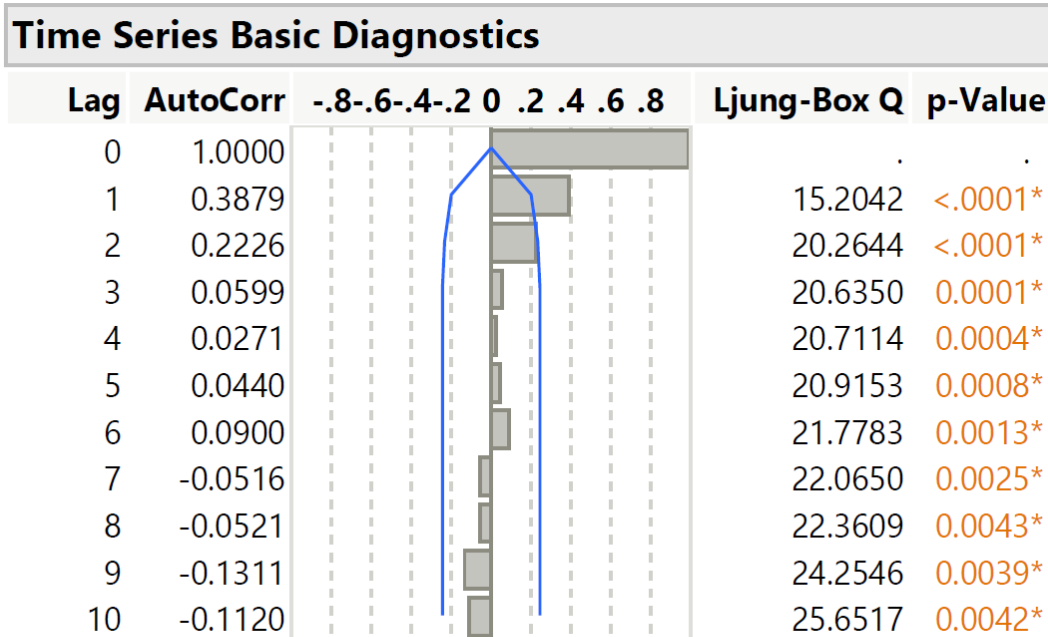
since $\bar{e}_{\cdot} = 0$

$$r_k = \hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0} = \frac{\sum_{t=1}^{N-k} e_t e_{t+k}}{\sum_{t=1}^N e_t^2} \quad k \leq N/4$$

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\text{Cov}(\varepsilon_t, \varepsilon_{t+k})}{\text{Var}(\varepsilon_t)}$$

Do we have some test procedure to quantitative check the “**no auto-correlation**” assumption? $H_0 : \rho_k = 0 \quad \forall k$

There are a lot of tests (**Bartlett**, **Liung-Box**, **Durbin Watson...**) but I suggest you use the **JMP Liung-Box test** (for the 98 data)



$$\left\{ \begin{array}{l} SE(r_1) = \sqrt{\frac{1}{N}} \quad \text{Simple TEST} \\ SE(r_k) = \sqrt{\frac{1 + 2 \sum_{m=1}^{k-1} r_m^2}{N}} \quad t_k = \frac{r_k}{SE(r_k)} \end{array} \right.$$

$$\text{if } |t_k| > t_{1-\alpha}(N-1)$$

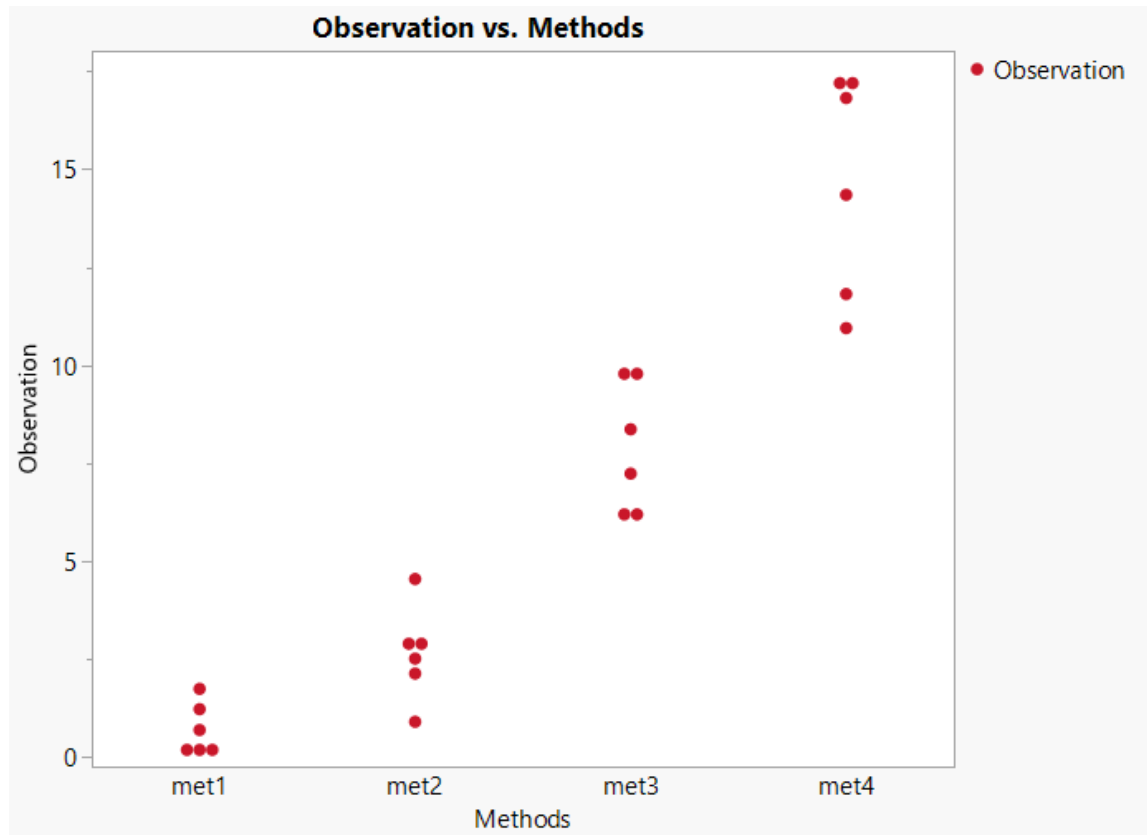
$$\text{if } Q(k) = N(N+2) \sum_{m=1,k} \left(\frac{r_m^2}{N-m} \right) > \chi_{1-\alpha}^2(k) \rightarrow \text{Reject } H_0$$

Interesting books on **Time series**

- Box G.E.P., Jenkins G.M., Reinsel G.C. «Time series Analysis», Wiley, 2008
- Priestley M.B. «Spectral analysis and Time series», Academic Press, 1981
- Econometrics Books

Different Variances

met1	met2	met3	met4
0.34	0.91	6.31	17.15
0.12	2.94	8.37	11.82
1.23	2.14	9.75	10.95
0.70	2.36	6.09	17.20
1.75	2.86	9.82	14.35
0.12	4.55	7.24	16.82



Test for Equal Variances

- If we have just two **Treatments** we can use the **Fisher test** on the **Variance ratio**.
- If we have **a** treatments (**a levels**) we can perform a **Fisher-Bonferroni Variance test**
- I suggest using more specialized tests.
 - **Hartley** (constant level replicates) **NO**
 - **Bartlett** (It strongly assumes data normality)
 - ➔ – **Brown&Forsythe** (a modification of **Levene's test**) (Robust, $n > 2$)

Brown&Forsythe's test (we don't prove it)

The **Brown&Forsythe test** (an updated Levene's test version) is robust to the assumption of normality.

The test is based on the difference between each **Studentized Residual SRES** and the corresponding Level SRES **Median** (**Levene** test used the **Mean**)

$\widetilde{SRES}_i$ = Median of SRES belonging to the population i

$SRES_{ij}^*$ = Transformed SRES $SRES_{ij}^* = \left| SRES_{ij} - \widetilde{SRES}_i \right|$

We apply the standard ANOVA Test on the **transformed** data

If there were different variances, the data would be spreaded differently according to their factor level

Example 3.7

1) Compute the **Parameter Estimates** of the Model

met1	0.710
met2	2.627
met3	7.930
met4	14.715

2) Compute the **Studentized Residuals** of the Model

met1	met2	met3	met4
-0.230052894	-1.067362528	-1.007258618	1.5139967502
-0.366841102	0.1948195681	0.2735764148	-1.800008457
0.3233175812	-0.302592095	1.1316115341	-2.340943641
-0.006217646	-0.165803888	-1.144046826	1.5450849792
0.6466351623	0.1450784018	1.1751350546	-0.226944071
-0.366841102	1.1958605406	-0.42901756	1.3088144391

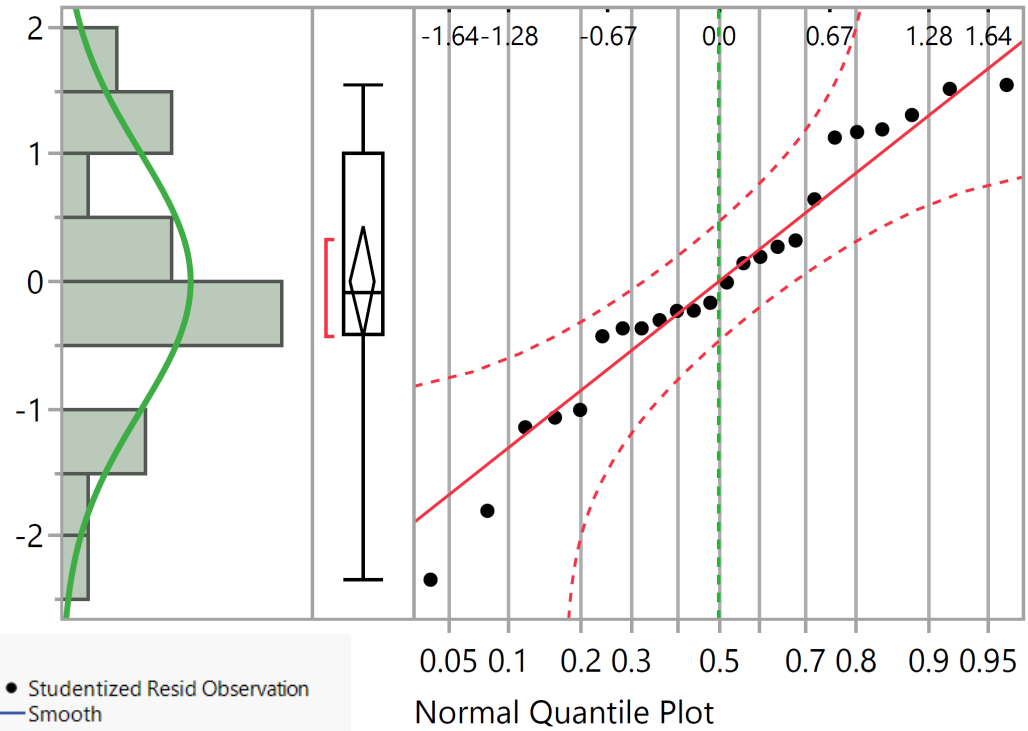
Pay attention: too many decimals. That is especially wrong in presentations

Studentized Resid Observation

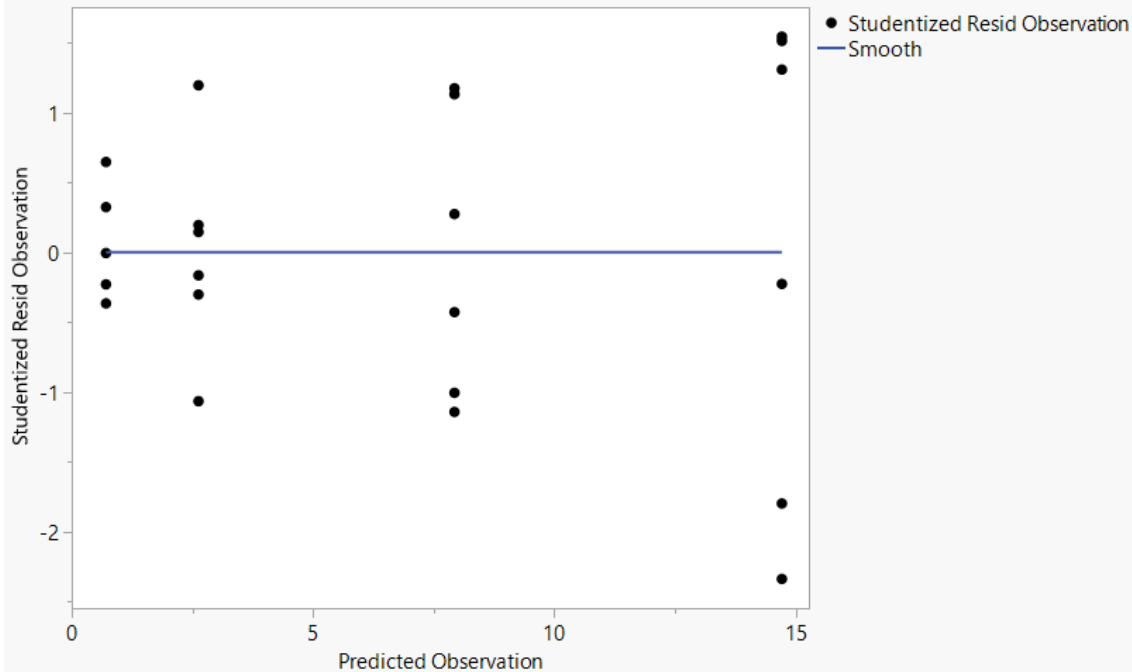
Goodness-of-Fit Test

	W	Prob<W
Shapiro-Wilk	0.9570144	0.3814
	A ²	Simulated p-Value
Anderson-Darling	0.3639055	0.4312

Note: Ho = The data is from the Normal distribution.
Small p-values reject Ho.

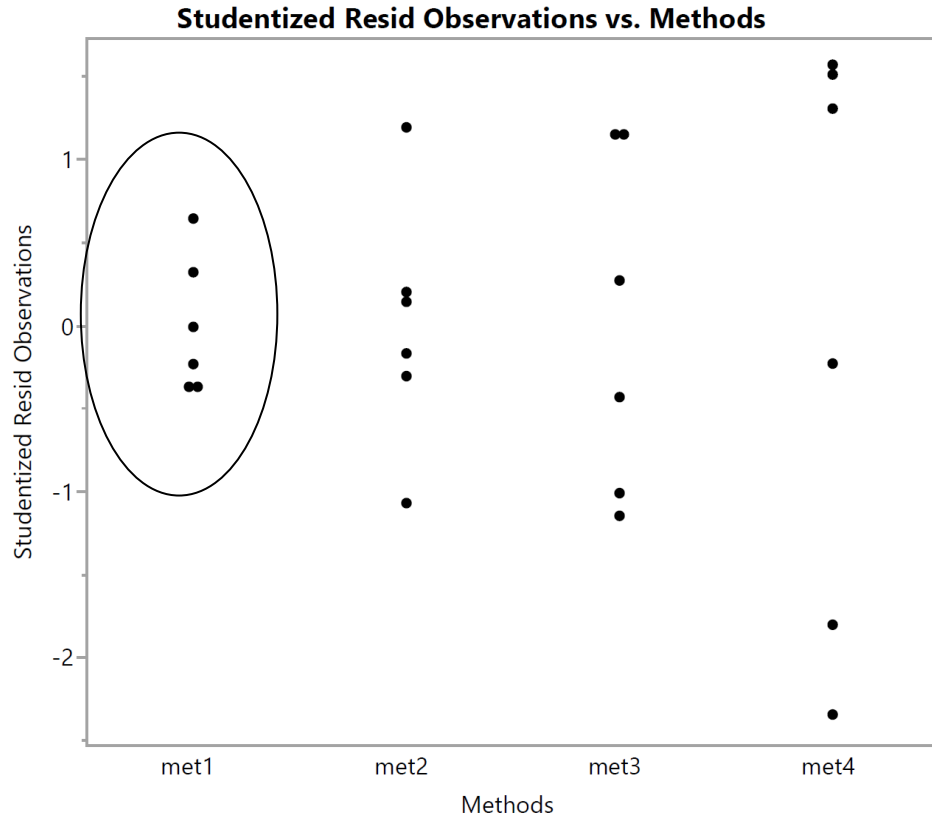
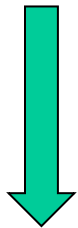


Studentized Resid Observation vs. Predicted Observation

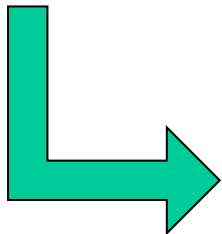


Test

SRES_1	Ord_SRES_1
-0.230053	-0.366841
-0.366841	-0.366841
0.323318	-0.230053
-0.006218	-0.006218
0.646635	0.323318
-0.366841	0.646635



$$\widetilde{SRES}_{met1} = \frac{-0.230053 - 0.006218}{2} = -0.1181355$$



$$SRES_{ij}^* = |SRES_{ij} - \widetilde{SRES}_i|$$



SRES*
 0.111918
 0.248706
 0.441453
 0.111918
 0.764770
 0.248706

Factor	Type	Levels	Values
level	Fixed	4	met 1; met 2; met 3; met 4

Analysis of Variance

Source	DF	Adj SS	Adj MS	F-Value	P-Value
level	3	4.473	1.4909	4.57	0.014
Error	20	6.527	0.3264		
Total	23	11.000			

JMP v17 Command (from command Fit Y by X)

Tests that the Variances are Equal

Test	F Ratio	DFNum	DFDen	Prob > F
O'Brien[.5]	5.3313	3	20	0.0073*
Brown-Forsythe	4.5684	3	20	0.0136*
Levene	6.4060	3	20	0.0032*
Bartlett	2.9986	3	.	0.0293*

Data Transformation

- When tests indicate a hypothesis violation (Variance or Normality) we could try to **transform** the **original data**
- This is mandatory if we want to make a **Post-Anova** analysis (i.e. a simultaneous comparison test)
- We will study two methods
 - **Taylor series expansion (Delta Method)**
 - **BOX-COX Transformation**

Taylor Series

$$V(Y) \equiv \sigma^2 = g(\mu)$$

- Let's suppose that

$$E(Y) = \mu$$

Transform Y to $Y^* = h(Y)$ $\ni$ $V(Y^*) = \text{const}$

$$Y^* = h(Y) = h(\mu) + h'(\mu)(Y - \mu) + \text{Remainder}$$

$$V(Y^*) = [h'(\mu)]^2 V(Y) = [h'(\mu)]^2 g(\mu) = \text{const} = 1$$

$$h(\mu) = \int \frac{1}{\sqrt{g(\mu)}} d\mu$$

$$\text{If } g(\mu) = k\mu \rightarrow h(\mu) = \int \frac{1}{\sqrt{k\mu}} d\mu = \frac{2}{\sqrt{k}} \sqrt{\mu} \rightarrow Y^* = \sqrt{Y}$$

Summing up

- Suppose that the $g(\mu)$ function (Variance vs level mean) is known or assumed
- Integrate e find the transformation function $h(y)$ near the mean
- Pay attention: the function $h(y)$ must be monotone

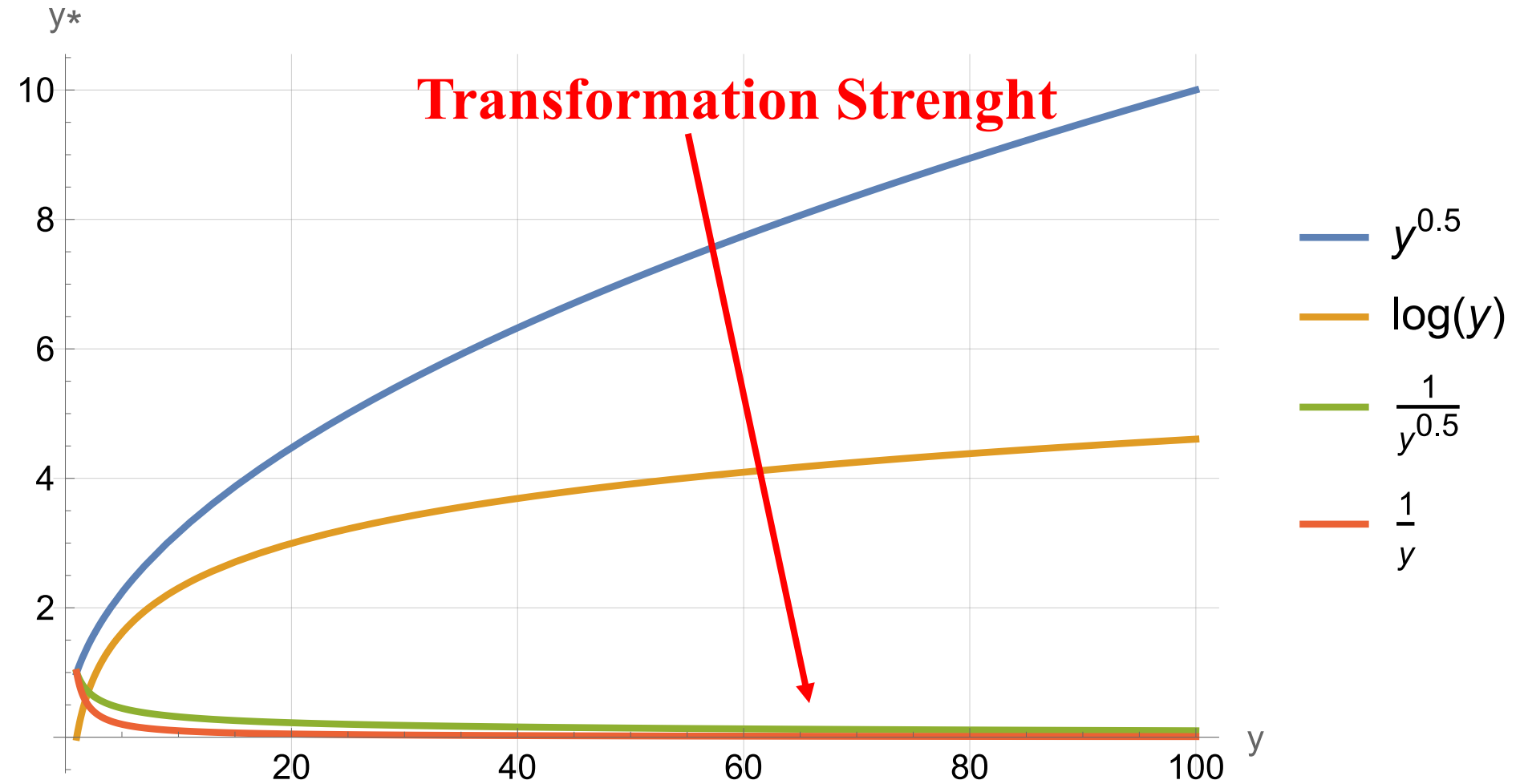
- A particular class of transformation: **Power Function**

$$\sigma_Y \propto \mu^\alpha \quad \rightarrow \quad y^* = y^\lambda$$

α	λ	Transformation
0	1	No transformation
1/2	1/2	$\sqrt{y}$
1	0	$\ln(y)$
3/2	-1/2	$1 / \sqrt{y}$
2	-1	$1 / y$

- The **strenght** of the transformation increases from up to down in the table

The **strenght** of the **transformation** increases with the reduction of the amplitude of the variation



Box-Cox Transformation (don't prove)

- This method is for **Power Transformations**
- Indexed by λ , the transformation is:

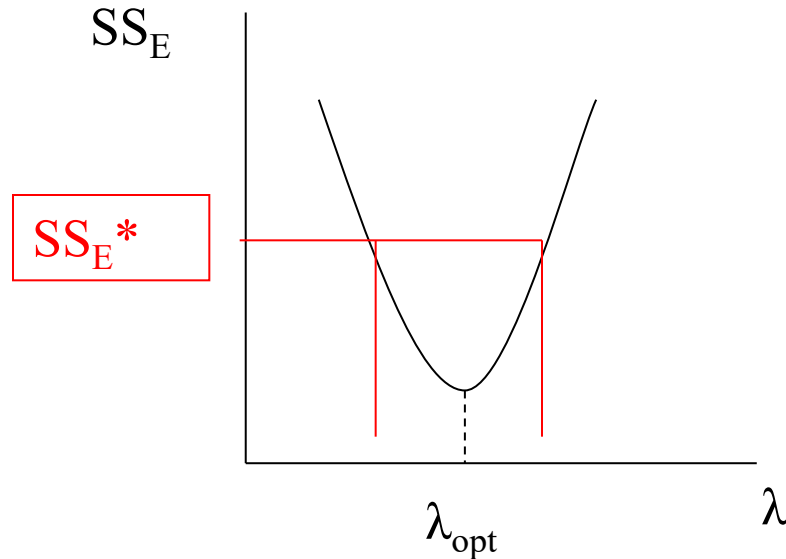
$$y^* = y_{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda \bar{y}^{\lambda-1}} & \text{for } \lambda \neq 0 \\ \bar{y} \ln y & \text{for } \lambda = 0 \end{cases} \quad \text{where } \bar{y} = \sqrt[N]{\prod_i y_i}$$

Geometric Mean

Note: the data should be positive

- For each λ compute $SS_E(\lambda)$ using the same tentative model

- Find the minimum of SS_E with respect to λ



- Compute the λ confidence interval

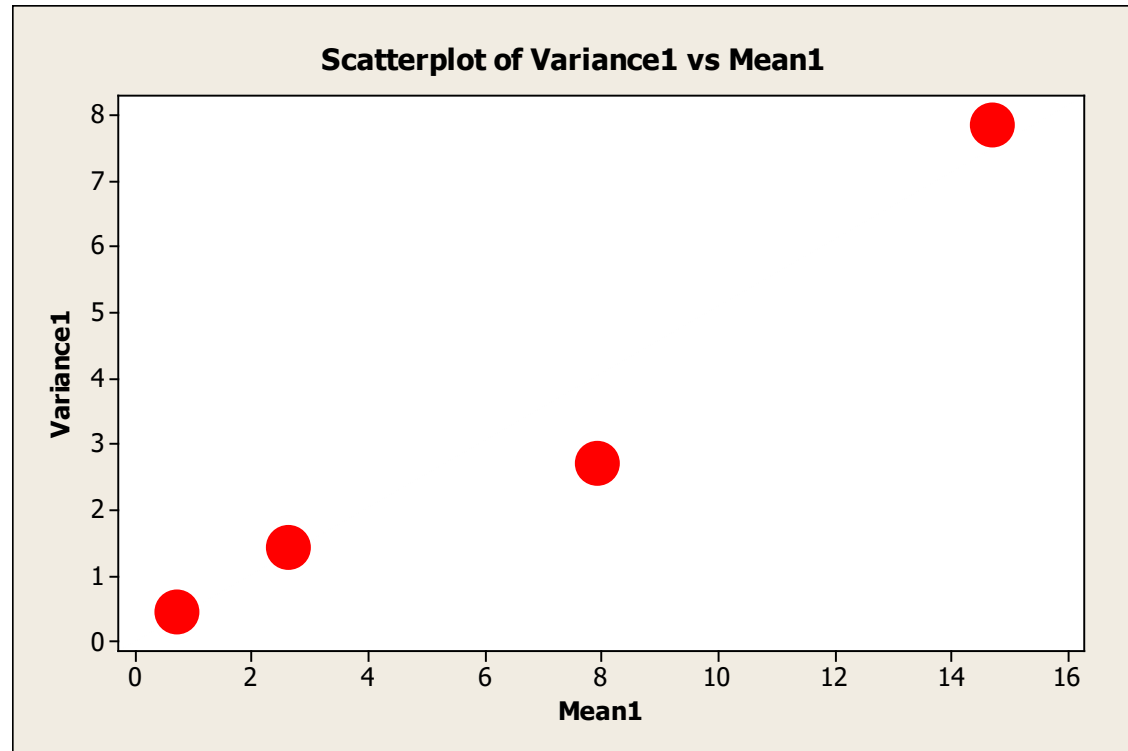
$$SS_E^* = SS_E(\lambda_{opt}) \left(1 + \frac{t_{\alpha/2}^2(df_E)}{df_E} \right)$$

- Exact formula is $\longrightarrow SS_E^* = SS_E(\lambda_{opt}) e^{\chi_{\alpha}^2(1)/N}$

Example: Taylor transformation

For each **met** level
compute:

- Sampling mean
- Sampling variance
- Graph Variance vs Mean



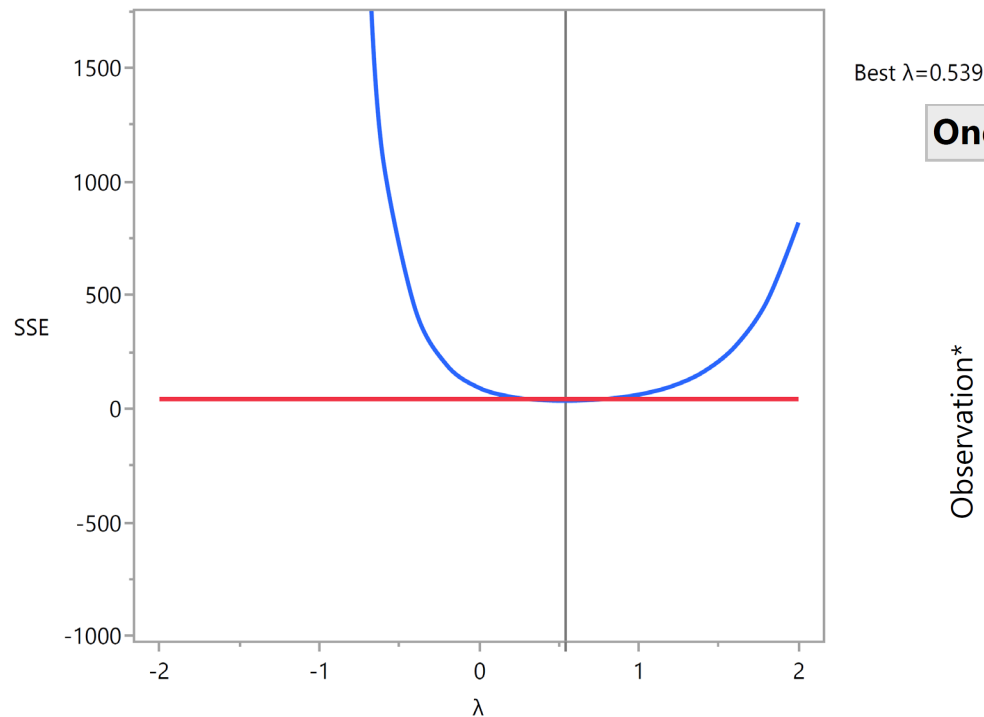
We suppose the relationship $V(Y) \equiv \sigma^2 = g(\mu)$
 $\sigma^2 = k\mu \rightarrow \sigma \propto \mu^{1/2}$

We know (see the transformation Table) that the corresponding transformation is the **Square Root**

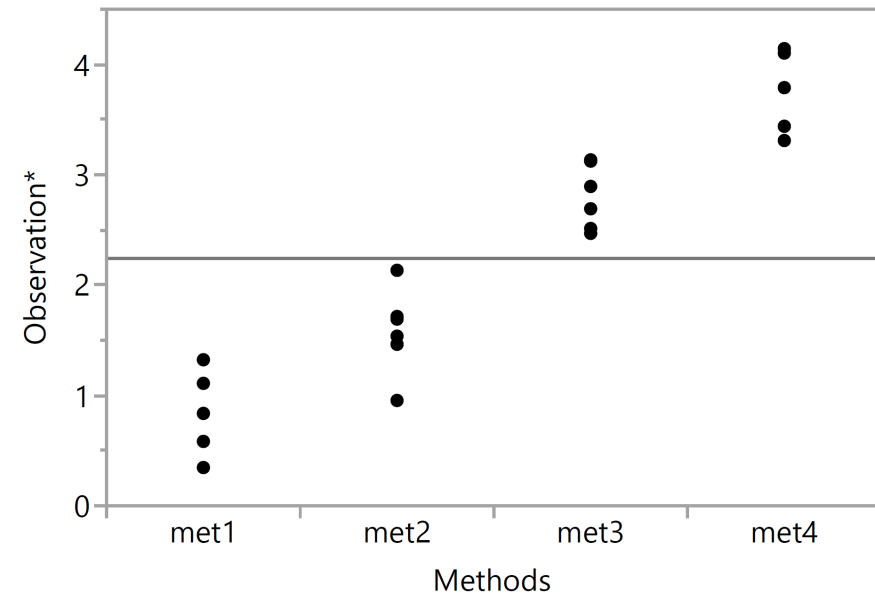
Box-Cox Transformation

JMP: Optimal Lambda= 0.539 but we select **Lambda=0.5**

Box-Cox Transformations



Oneway Analysis of Observation* By Methods



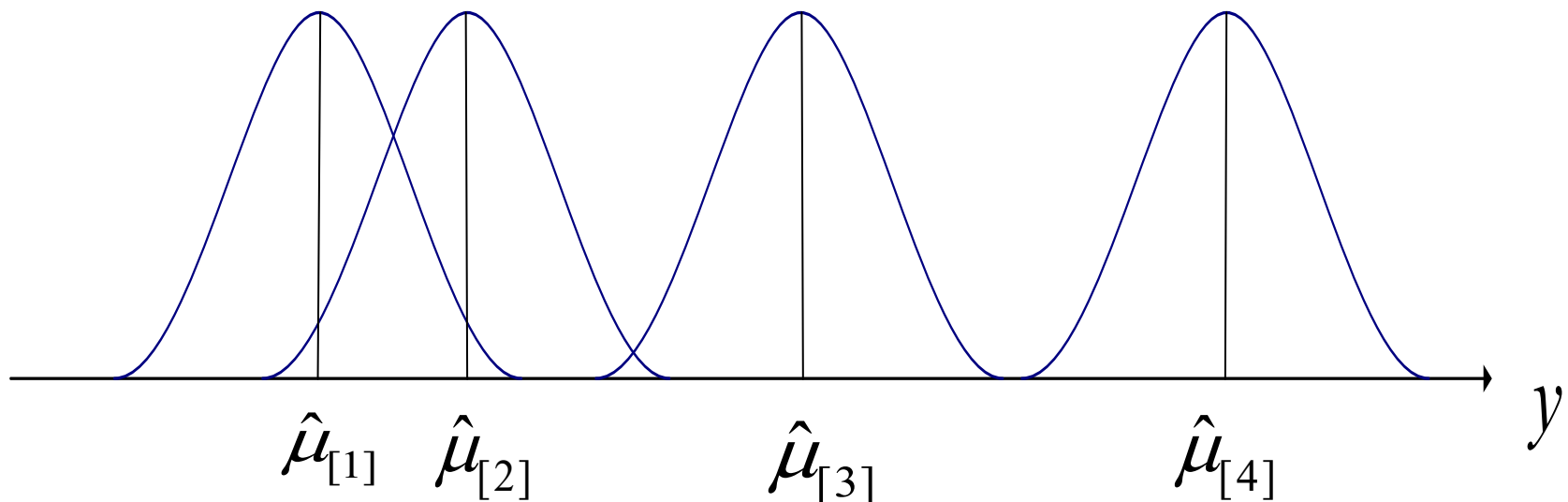
Let us do the analysis again and the problem disappeared

Two examples

- Selecting the best **Cutting Tool**
 - Heat resistant alloys (e.g. INCONEL) are used for aircraft engines
 - Four suppliers (SANDVIK, KENNAMETAL, MITSUBISHI and SECO) propose their best carbide inserts
 - We are interested in maximizing the **Tool Life**
 - Discuss the Experiment
- Selecting the best **Dispatching Rule**
 - In a Job Shop can be useful to have a dispatching rule to schedule each machine buffer
 - We consider SPT, FIFO, LIFO and EDD (Earliest Due Date)
 - We are interested in minimizing the WIP (Work In Process)
 - Discuss the Experiment

Comparison among Treatment Means (Post-ANOVA Analysis)

- **ANOVA**, test H_0 : all the treatment means are equal
- If we **reject** the null hypothesis, we don't know **which** treatment **means** are statistically different
- We should perform a **Multiple Comparison Test of Means**



Comparisons among the Level Means (Treatments Means)

- We introduce the concept of **Contrast L**

$$L = \sum_i c_i \mu_i \quad \text{with} \quad \sum_i c_i = 0$$

- i.e.
 - a) Difference between two means $L_{12} = \mu_1 - \mu_2$
 $c_1 = 1, c_2 = -1, c_3 = 0 \dots c_a = 0$
 - b) Test if $H_0 : \mu_1 + \mu_2 = \mu_3 + \mu_4$

$$L = \mu_1 + \mu_2 - \mu_3 - \mu_4 \quad H_0 : L = 0 \quad \text{vs} \quad H_1 : L \neq 0$$

- Contrast **estimate**

$$\hat{L} = \sum_i c_i \hat{\mu}_i = \sum_i c_i \bar{y}_i.$$

- Statistical properties
(independent data)

$$E(\hat{L}) = \sum_i c_i E(\bar{y}_{i.}) = \sum_i c_i \mu_i = L$$

$$V(\hat{L}) = \sum_i c_i^2 V(\bar{y}_{i.}) = \sigma^2 \sum_i \frac{c_i^2}{n_i}$$

if data are Normal or using the Central Limit Theorem

$$\hat{L} \sim N\left(L, \sigma^2 \sum_i c_i^2 / n_i\right) \rightarrow \frac{\hat{L} - L}{\sigma \sqrt{\sum_i c_i^2 / n_i}} \sim N(0, 1)$$

If $\sigma^2 \leftarrow MS_E \rightarrow \frac{\hat{L} - L}{\sqrt{MS_E \sum_i c_i^2 / n_i}} \sim t(df_E)$

Discuss CI and Test

If we are interested in a **specific contrast**

$$L \equiv D_{ij} = \mu_i - \mu_j \rightarrow \hat{L} \equiv \hat{D}_{ij} = \hat{\mu}_i - \hat{\mu}_j$$

$$\hat{D}_{ij} \sim N\left(D_{ij}, \sigma^2 \left(\frac{1}{n_i} + \frac{1}{n_j}\right)\right) \rightarrow \frac{\hat{D}_{ij} - D_{ij}}{\sigma \sqrt{\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}} \sim N(0,1)$$

$$\text{Reject } H_0 : D_{ij} = 0 \text{ if } \frac{|\hat{D}_{ij}|}{\sigma \sqrt{\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}} > z_{\alpha/2}$$

$$\text{If } \sigma^2 \leftarrow MS_E \rightarrow \frac{\hat{D}_{ij} - D_{ij}}{\sqrt{MS_E \left(\frac{1}{n_i} + \frac{1}{n_j}\right)}} \sim t(df_E)$$

$$\text{Reject } H_0 : D_{ij} = 0 \text{ if } \frac{|\hat{D}_{ij}|}{\sqrt{MS_E \left(\frac{1}{n_i} + \frac{1}{n_j}\right)}} > t_{\alpha/2}(df_E)$$

Usually $n_i = n_j = n$

If we had chosen two **specific** factor levels **before experimenting** we could use the previous formulas

However to compare **all** treatment **mean pairs** we should consider the family of **r** mean pairs with

$$r = \frac{a(a-1)}{2}$$

Two treatment means will be different (in a statistical sense) with a **Familywise** error α_{FAM} if:

$$\frac{|\hat{D}_{ij}|}{\sqrt{\hat{V}(\hat{D}_{ij})}} > K_{\alpha_{FAM}} \text{ or } |\hat{D}_{ij}| > K_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{ij})} \Rightarrow \text{Reject } H_0 : D_{ij} = 0$$

How to compute **K** ? The method with **Minimum K** is **the best!**

- There are a lot of Methods
 - **t-Bonferroni**
 - **Scheffé**
 - **Tukey**
 - **Fisher LSD**
 - **Dunnet**
 - (Graphical Method), (Duncan), (Newman-Keuls, ...)

t-Bonferroni Method

$$\frac{\alpha}{2} \leftarrow \frac{\alpha_{FAM}}{2r}$$

$$K_{\alpha_{FAM}} \equiv B_{\alpha_{FAM}} = t_{\alpha_{FAM}/2r}(df_E)$$

$$\text{Reject } H_0 : D_{ij} = 0 \quad \text{IF} \quad \frac{|\hat{D}_{ij}|}{\sqrt{\hat{V}(\hat{D}_{ij})}} > B_{\alpha_{FAM}}$$

We can also compute the CI

$$\hat{D}_{ij} - B_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{ij})} \leq D_{ij} \leq \hat{D}_{ij} + B_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{ij})}$$

- Remind the Hypotheses:
 - Normal,
 - Independent,
 - Constant variance.
- The method, with some minor changes, can be used **ALSO** to compare *a* treatment Mean pairs when the variances are **different** (known or estimated)

How ? Try it yourself

Scheffé Method

- Hypotheses: **as usual**
- It considers the family of **all possible contrasts**:

$$L = \sum_i c_i \mu_i \quad \text{con} \quad \sum_i c_i = 0$$

- For the **Test/Confidence Interval** for all possible contrasts with ***Family Confidence Coefficient*** $(1-\alpha_{FAM})$

$$S_{\alpha_{FAM}}^2 = (a-1)F_{\alpha}(a-1, df_E)$$

$$\text{Test: } \mathbf{Reject} \ L=0 \text{ if } \frac{|\hat{L}|}{\sqrt{\hat{V}(\hat{L})}} > S_{\alpha_{FAM}}$$

$$\text{CI: } \hat{L} - S_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{L})} \leq L \leq \hat{L} + S_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{L})}$$

Tukey Method

- Hypotheses:
 - Normal, independent , constant variance
 - Treatment replicates are **constant**. (Tukey-Kramer proved it for variable **n**)
 - For **all pairs of treatment means** **$a(a-1)/2$**

$$(\bar{y}_{1\cdot} - \mu_1), (\bar{y}_{2\cdot} - \mu_2), \dots, (\bar{y}_{i\cdot} - \mu_i), \dots, (\bar{y}_{a\cdot} - \mu_a)$$

- The single quantities are:
 - Random variable
 - Independent
 - Normal , with null expected value and variance **σ^2/n** or **MS_E/n** with **$a(n-1)=N-a$** *df*

Sample of **a**
observations

$$\frac{\text{Max}_i (\bar{y}_{i\cdot} - \mu_i) - \min_i (\bar{y}_{i\cdot} - \mu_i)}{\sqrt{MS_E / n}} \sim q(a, df_E)$$

Studentized Range Distribution Tab VII

• and

$$\Pr \left\{ \frac{\text{Max}_i (\bar{y}_{i\cdot} - \mu_i) - \min_i (\bar{y}_{i\cdot} - \mu_i)}{\sqrt{MS_E / n}} \leq q_{\alpha_{FAM}}(a, df_E) \right\} = 1 - \alpha_{FAM}$$

For each pair f e l

$$\left| (\bar{y}_{f\cdot} - \mu_f) - (\bar{y}_{l\cdot} - \mu_l) \right| \leq \text{Max}_i (\bar{y}_{i\cdot} - \mu_i) - \min_i (\bar{y}_{i\cdot} - \mu_i)$$

$$\Pr \left\{ \frac{\left| (\bar{y}_{f\cdot} - \mu_f) - (\bar{y}_{l\cdot} - \mu_l) \right|}{\sqrt{MS_E / n}} \leq q_{\alpha_{FAM}}(a, df_E) \right\} \geq 1 - \alpha_{FAM}$$

$$\Pr \left\{ \frac{\left| \left(\bar{y}_{f\cdot} - \bar{y}_{l\cdot} \right) - \left(\mu_f - \mu_l \right) \right|}{\sqrt{MS_E / n}} \leq q_{\alpha_{FAM}} \left(a, df_E \right) \right\} \geq 1 - \alpha_{FAM}$$

$$D_{fl} = \mu_f - \mu_l = \tau_f - \tau_l \quad \hat{D}_{fl} = \hat{\mu}_f - \hat{\mu}_l = \bar{y}_{f\cdot} - \bar{y}_{l\cdot} \quad \hat{V} \left(\hat{D}_{fl} \right) = 2 \frac{MS_E}{n}$$

$$\Pr \left\{ \frac{\left| \hat{D}_{fl} - D_{fl} \right|}{\sqrt{MS_E / n}} \leq q_{\alpha_{FAM}} \left(a, df_E \right) \right\} \geq 1 - \alpha_{FAM}$$

Divide both terms of the inequality by $\sqrt{2}$

$$\text{Reject } H_0 : D_{fl} = 0 \text{ if } \frac{\left| \hat{D}_{fl} \right|}{\sqrt{\hat{V} \left(\hat{D}_{fl} \right)}} > T_{\alpha_{FAM}}$$

$$\hat{D}_{fl} - T_{\alpha_{FAM}} \sqrt{\hat{V} \left(\hat{D}_{fl} \right)} \leq D_{fl} \leq \hat{D}_{fl} + T_{\alpha_{FAM}} \sqrt{\hat{V} \left(\hat{D}_{fl} \right)}$$

JMP

$$\frac{1}{\sqrt{2}} q_{\alpha_{FAM}} \left(a, df_E \right) = T_{\alpha_{FAM}}$$

- **TUKEY-KRAMER** proved that in case of **variable n**:

$$D_{fl} = \mu_f - \mu_l = \tau_f - \tau_l$$

$$\hat{D}_{fl} = \hat{\mu}_f - \hat{\mu}_l = \bar{y}_{f\cdot} - \bar{y}_{l\cdot}$$

$$\hat{V}(\hat{D}_{fl}) = MS_E \left(\frac{1}{n_f} + \frac{1}{n_l} \right)$$

$$\frac{|\hat{D}_{fl}|}{\sqrt{MS_E \left(\frac{1}{n_f} + \frac{1}{n_l} \right)}} > T_{\alpha_{FAM}} \quad \text{with} \quad T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(a, df_E)$$

$$\hat{D}_{fl} - T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{fl})} \leq D_{fl} \leq \hat{D}_{fl} + T_{\alpha_{FAM}} \sqrt{\hat{V}(\hat{D}_{fl})}$$

Comparison of Methods

Reject if $\frac{|\hat{D}_{fl}|}{\sqrt{MS_E \left(\frac{1}{n_f} + \frac{1}{n_l} \right)}} > K_{\alpha_{FAM}}$ where $K_{\alpha_{FAM}} = \begin{cases} B_{\alpha_{FAM}} \\ S_{\alpha_{FAM}} \\ T_{\alpha_{FAM}} \end{cases}$

$$B_{\alpha_{FAM}} = t_{\frac{\alpha_{FAM}}{2r}}(df_E)$$

$$S_{\alpha_{FAM}} = \sqrt{(a-1)F_{\alpha_{FAM}}(a-1, df_E)}$$

$$T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(a, df_E)$$

- In our **cotton%** example with *alfa*= 1%

$$B_{\alpha_{FAM}} = t_{\frac{\alpha}{2r}}(df_E) = t_{\frac{0.01}{2*10}}(20) = t_{0.0005}(20) = 3.8495$$

$$S_{\alpha_{FAM}} = \sqrt{(a-1)F_{\alpha}(a-1, df_E)} = \sqrt{4F_{0.01}(4, 20)} = \sqrt{4 * 4.4307} = 4.2098$$

$$T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha}(a, df_E) = \frac{1}{\sqrt{2}} q_{0.01}(5, 20) = \frac{5.29}{\sqrt{2}} = 3.7406$$

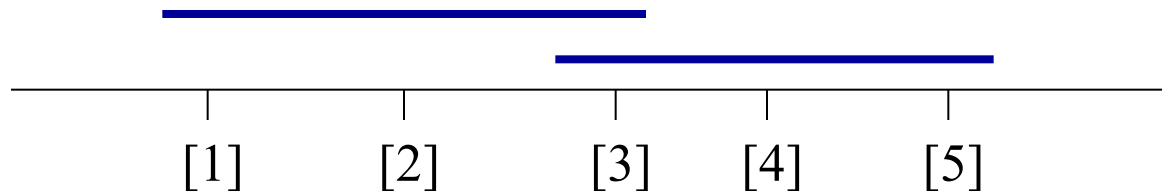
- We use the **TUKEY method** (minimum constant)
- All **pairs** of treatment means are statistical different if:

$$|\hat{D}_{fl}| > K \sqrt{\frac{2}{n} MS_E} \rightarrow |\hat{D}_{fl}| > 3.7406 \sqrt{\frac{2}{5} 8.06} \rightarrow |\hat{D}_{fl}| > 6.72$$

pair (f, l) is significant if $|\hat{D}_{fl}| > 6.72$

- Difference table (Cotton example, 5 levels)**

	[2]=5 10.8	[3]=2 15.4	[4]=3 17.6	[5]=4 21.6
[1]=1 → 9.8	1.0	5.6	7.8	11.8
[2]=5 → 10.8		4.6	6.8	10.8
[3]=2 → 15.4			2.2	6.2
[4]=3 → 17.6				4



JMP v17 Output

Command: from **Fit Y by X** remember to set the alpha level at 0.01 in this case

Confidence Quantile	
q*	Alpha
3.74275	0.01

$$q^* = T_{\alpha_{FAM}} = \frac{1}{\sqrt{2}} q_{\alpha_{FAM}}(a, df_E)$$

All Pairwise Differences Connecting Letters

Cotton Weight Percentage		Least Squares Mean
30	A	21.600000
25	A	17.600000
20	A B	15.400000
35	B	10.800000
15	B	9.800000

Levels not connected by same letter are significantly different.

Other comparison techniques

FISHER LSD - Least Significant Difference Test

- It is a classical *t-test* on all treatment mean pairs without the Bonferroni approach
- Two treatment means are different if

$$\left| \hat{D}_{fl} \right| > t_{\alpha/2}(df_E) \sqrt{MS_E \left(\frac{1}{n_f} + \frac{1}{n_l} \right)}$$

Dunnet Test

- Used to compare all treatment means with a **control treatment** (i.e. the current process condition).
- Only ***a-1*** comparisons

$$\left. \begin{array}{l} H_0 : \mu_f = \mu_a \\ H_1 : \mu_f \neq \mu_a \end{array} \right\} \text{ with } f = 1, 2, \dots, (a-1)$$

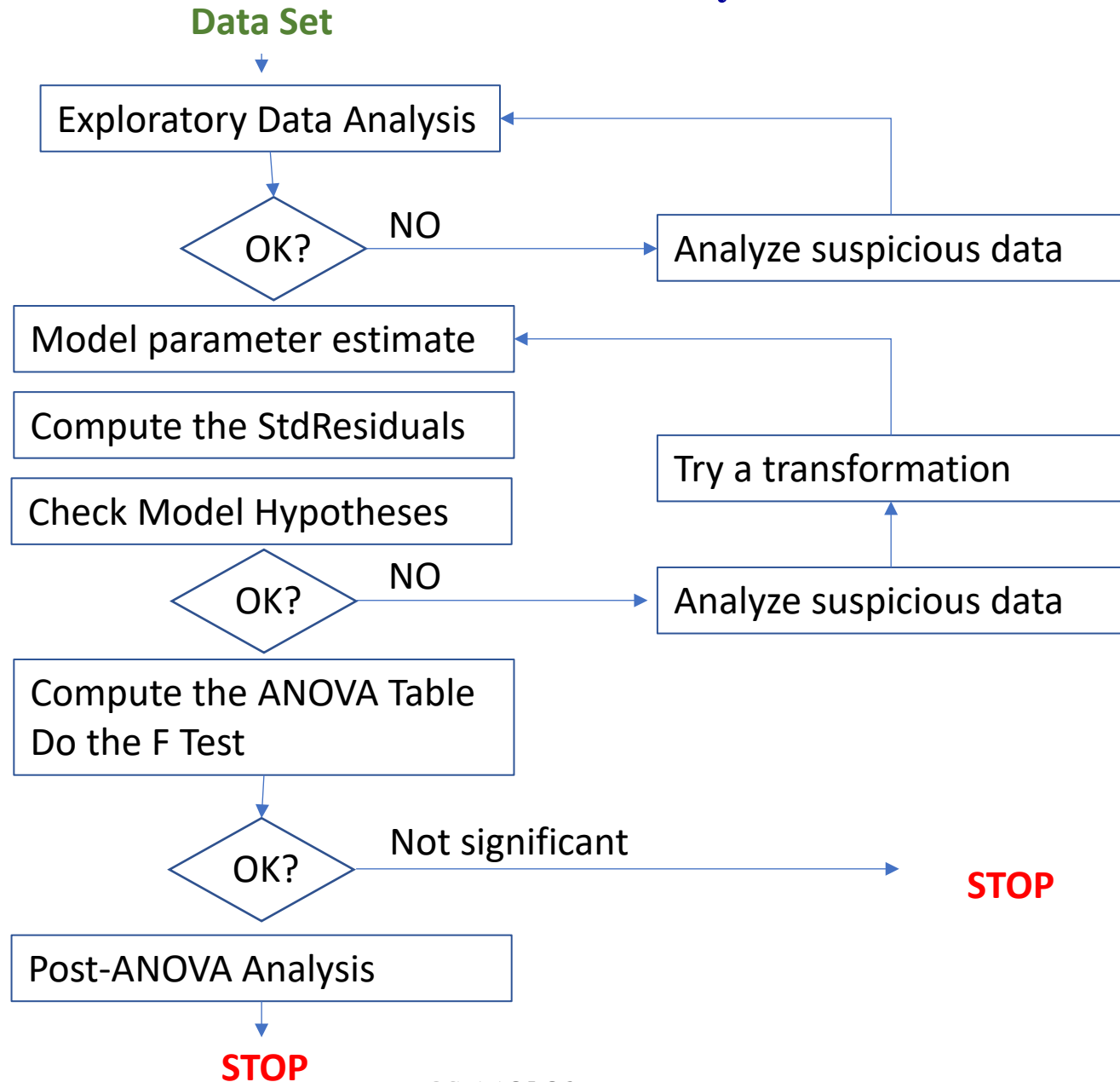
- The null hypothesis is rejected if

$$|\bar{y}_{f\cdot} - \bar{y}_{a\cdot}| > d_{\alpha_{FAM}}(a-1, df_E) \sqrt{MS_E \left(\frac{1}{n_f} + \frac{1}{n_a} \right)}$$

- Better if $\frac{n_a}{n_f} \simeq \sqrt{a}$

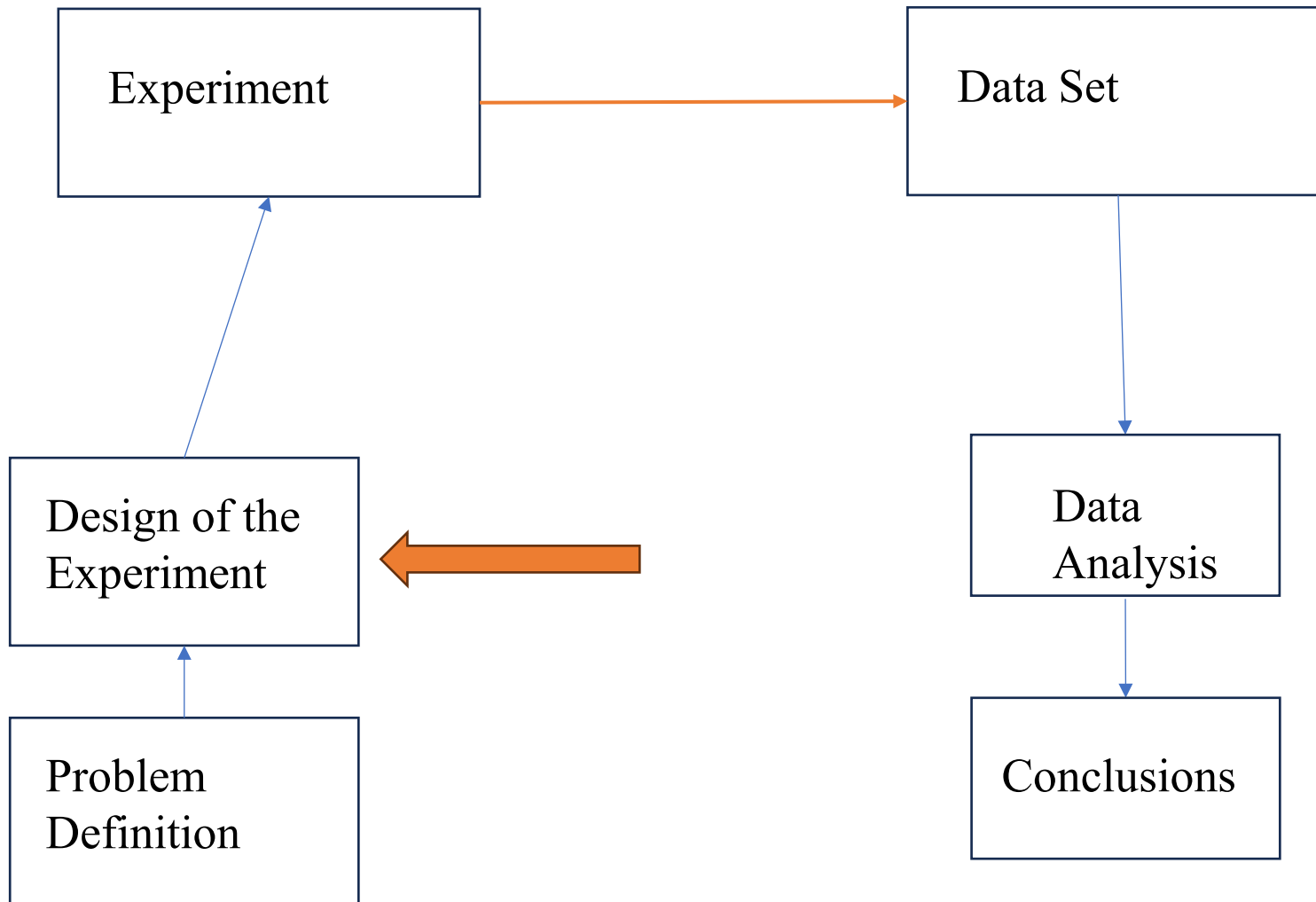
Tab VIII or JMP

Data Analysis WorkFlow



Global WorkFlow

- We will discuss the other side of an Experiment: the **Design**



Power of ANOVA F-Test

- If the null hypothesis H_0 is true

$$E(MS_{FAC}) = \sigma^2$$

$$\left(\frac{MS_{FAC}}{MS_E} \middle| H_0 \right) \sim F(df_{FAC}, df_E)$$

- If the alternative hypothesis H_1 is true

$$E(MS_{FAC}) = \sigma^2 + \Delta_{FAC}^2$$

$$\Delta_{FAC}^2 = \frac{n}{a-1} \sum_i \tau_i^2$$

$$\left(\frac{MS_{FAC}}{MS_E} \middle| H_1 \right) \sim F(df_{FAC}, df_E, \delta_{FAC})$$

$$\delta_{FAC} = \frac{\Delta_{FAC}^2}{\sigma^2} df_{FAC} = n \frac{\sum_{i=1,a} \tau_i^2}{\sigma^2}$$

**Non centrality
parameter**

- Fisher distribution and the non centrality parameter δ_{FAC}**

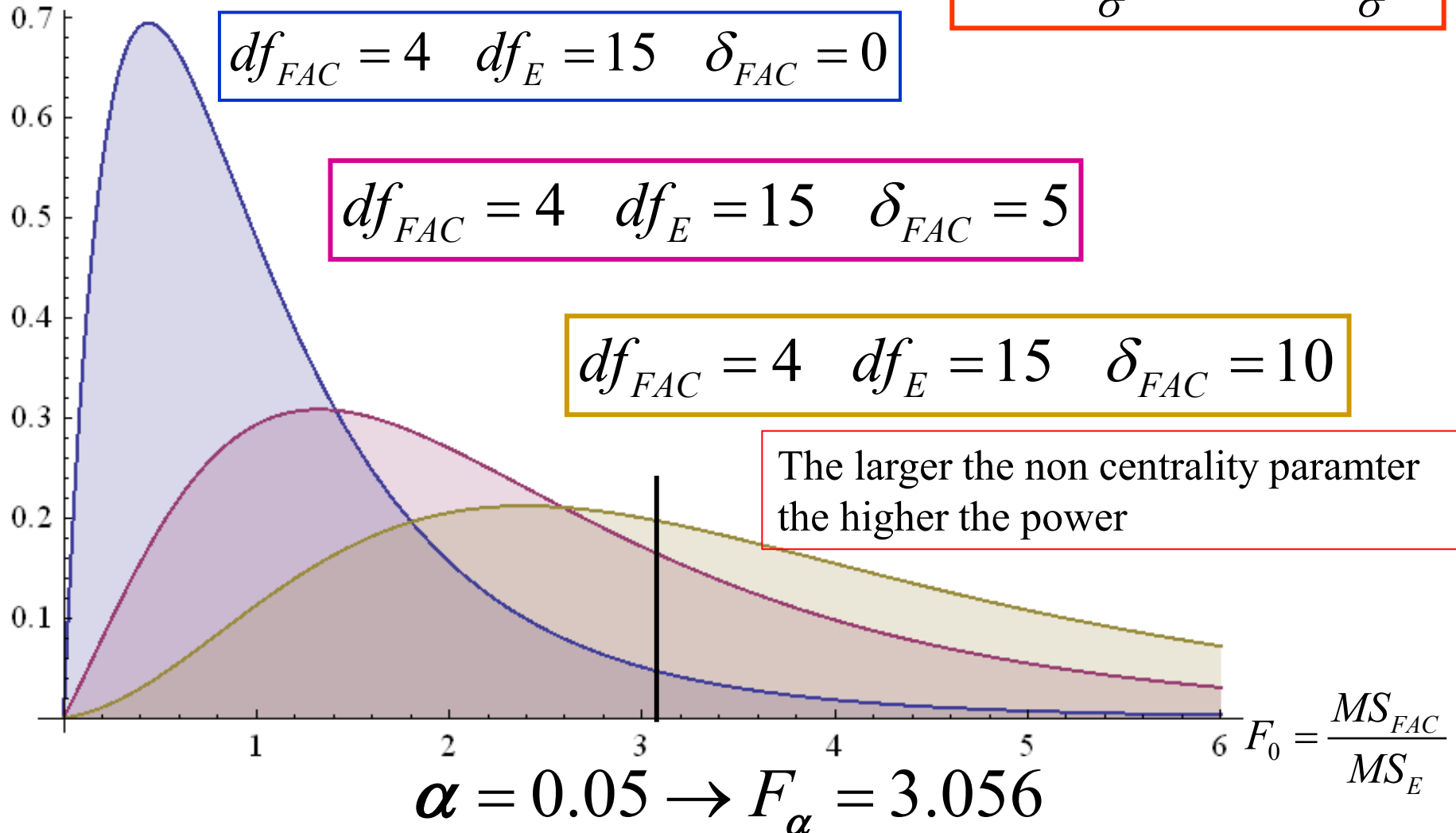
$$\delta_{FAC} = \frac{\Delta_{FAC}^2}{\sigma^2} df_{FAC} = n \frac{\sum_i \tau_i^2}{\sigma^2}$$

$$df_{FAC} = 4 \quad df_E = 15 \quad \delta_{FAC} = 0$$

$$df_{FAC} = 4 \quad df_E = 15 \quad \delta_{FAC} = 5$$

$$df_{FAC} = 4 \quad df_E = 15 \quad \delta_{FAC} = 10$$

The larger the non centrality parameter the higher the power



Statistical test errors

$$H_0 : \tau_i = 0 \quad \forall i \qquad H_1 : \exists (i, j) \ni \tau_i \neq \tau_j \neq 0$$

$$\alpha = \text{Prob} \left\{ F(df_{FAC}, df_E) > F_\alpha(df_{FAC}, df_E) | H_0 \right\}$$

$$\text{Power} = 1 - \beta = \text{Prob} \left\{ F(df_{FAC}, df_E, \delta_{FAC}) > F_\alpha(df_{FAC}, df_E) | H_1 \right\}$$

$$df_{FAC} = a - 1$$

$$df_E = a(n - 1)$$

$$\delta_{FAC} = n \frac{\sum_{i=1,a} \tau_i^2}{\sigma^2}$$

The Power depends on

$$\left\{ \begin{array}{c} \alpha \\ \sum_{i=1,a} \tau_i^2 / \sigma^2 \\ n \\ a \end{array} \right.$$

- Example **Direct Computational Method**

$$a = 4 \quad n = 3 \quad \delta_{FAC} = 36 \quad \alpha = 0.05$$

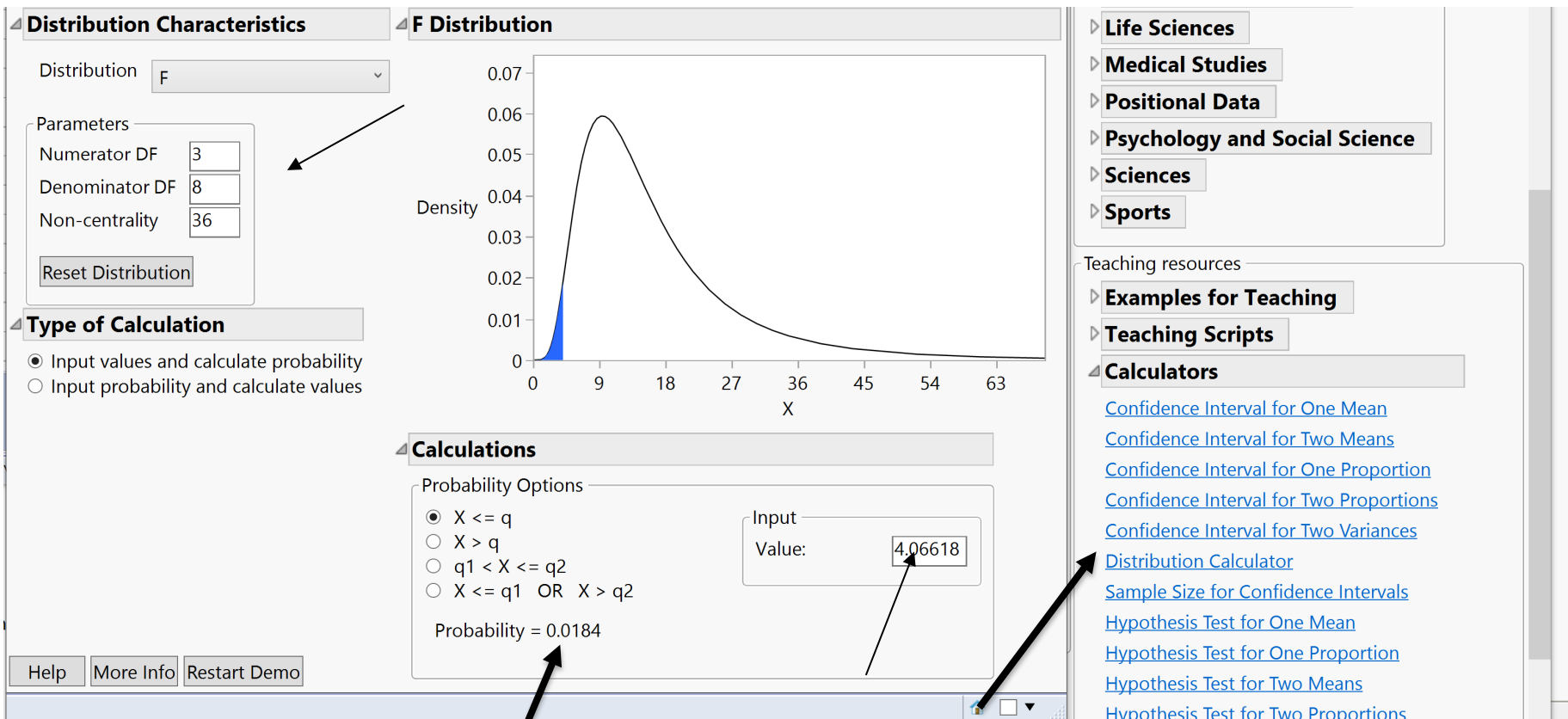
$$df_{FAC} = a - 1 = 4 - 1 = 3 \quad df_E = a(n - 1) = 4(3 - 1) = 8$$

$$F_{\alpha}(df_{FAC}, df_E) = F_{0.05}(3, 8) = 4.06618$$

$$\begin{aligned} \beta &= \text{Prob} \{ F(df_{FAC}, df_E, \delta_{FAC}) \leq F_{\alpha}(df_{FAC}, df_E) \} = \\ &= \text{Prob} \{ F(3, 8, 36) \leq 4.06618 \} = 0.018417 \end{aligned}$$

$$\text{Power} = 1 - \beta = 1 - 0.018417 = 0.98153$$

JMP v17 Help → Sample Index → Teaching Resources: Calculators → Distribution Calculator

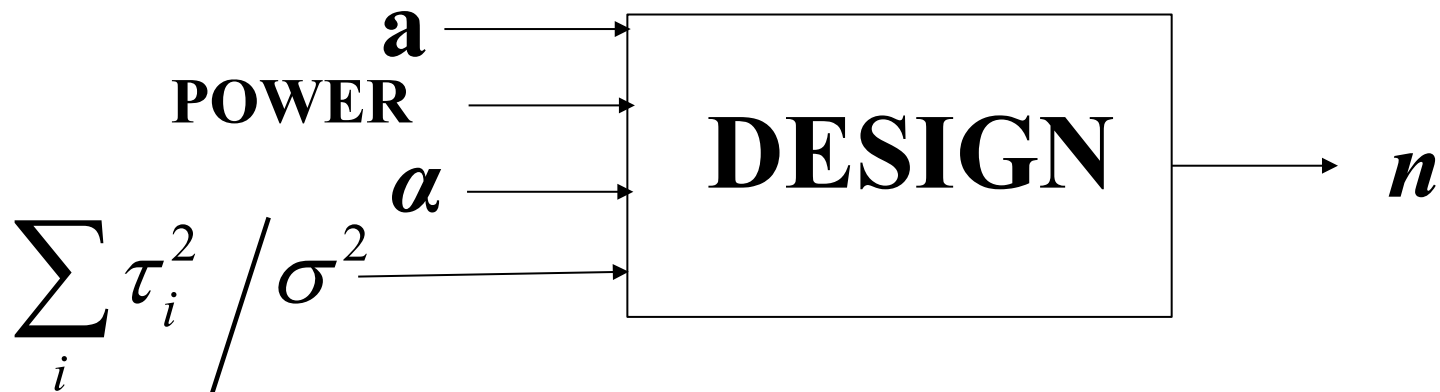


β

$$\text{Power} = 1 - \beta = 1 - 0.0184 = 0.9816$$

You can compute the **Power** just selecting **X>q** in the dialog windows

Design the Experiment: find (a, n)



$$\delta_{FAC} = n \frac{\sum_i \tau_i^2}{\sigma^2}$$

A) Define several sets of τ_i and choose the **minimum** value for $\sum_i \tau_i^2$ **but** we should have a prior estimate of σ

B) Minimum difference of interest d

- Let's define the minimum difference of interest d between any two means $\forall(i, j) \rightarrow \mu_i - \mu_j \leq d$

- The minimum $\sum_i \tau_i^2$ is attained when

$$\tau_i = 0 \quad \forall i \neq j, j' \quad \text{e} \quad \tau_j = -\frac{d}{2} \quad \tau_{j'} = \frac{d}{2} \quad \sum_i \tau_i^2 = \frac{d^2}{2}$$

$$\delta_{FAC} = n \frac{\sum_i \tau_i^2}{\sigma^2} \quad \delta_{FAC} = \frac{n}{2} \left(\frac{d}{\sigma} \right)^2$$

- Solution by trial and error (or math programming)

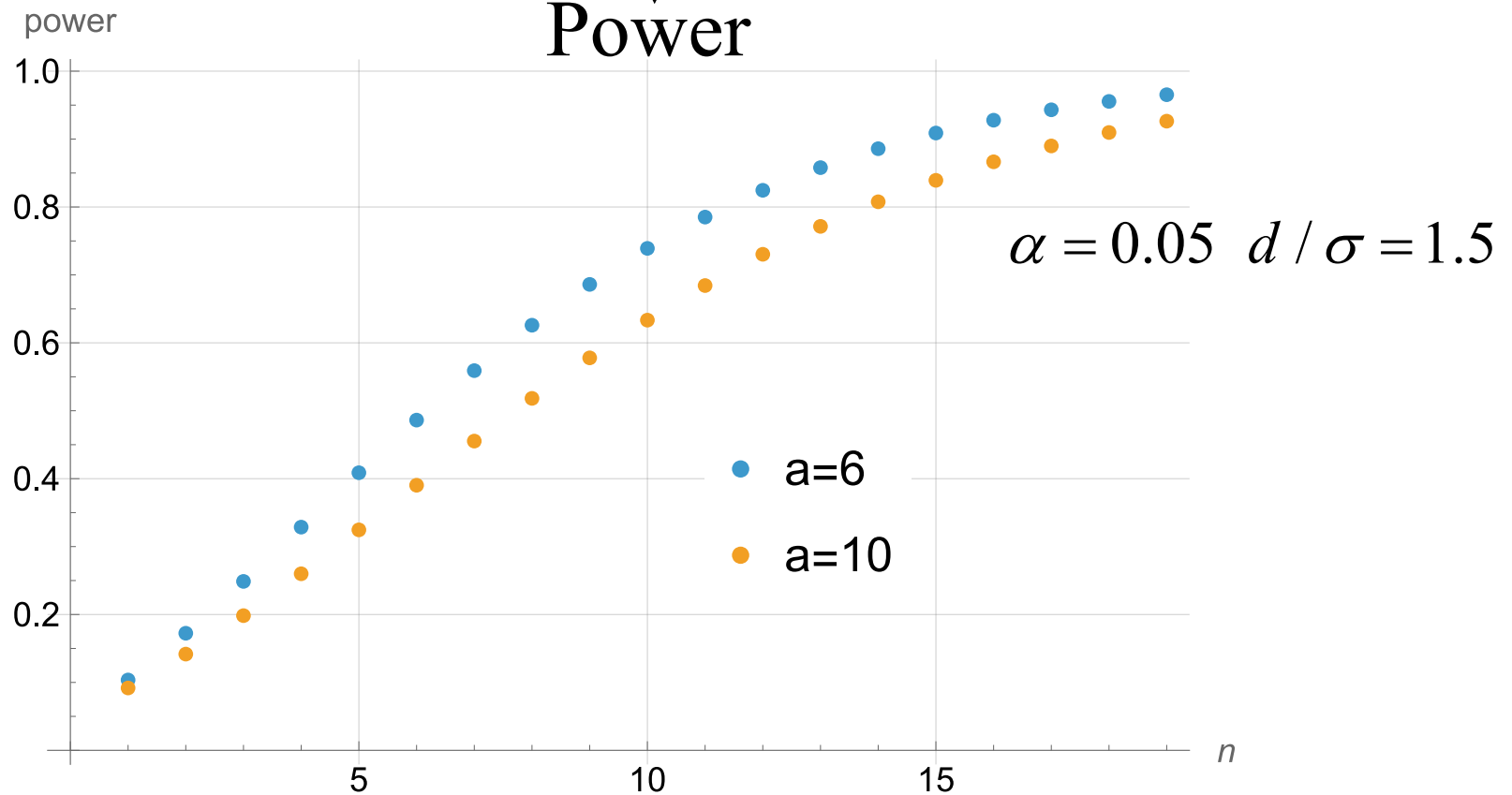
Computation flow

$$\boxed{\forall n \quad a \quad \alpha}$$

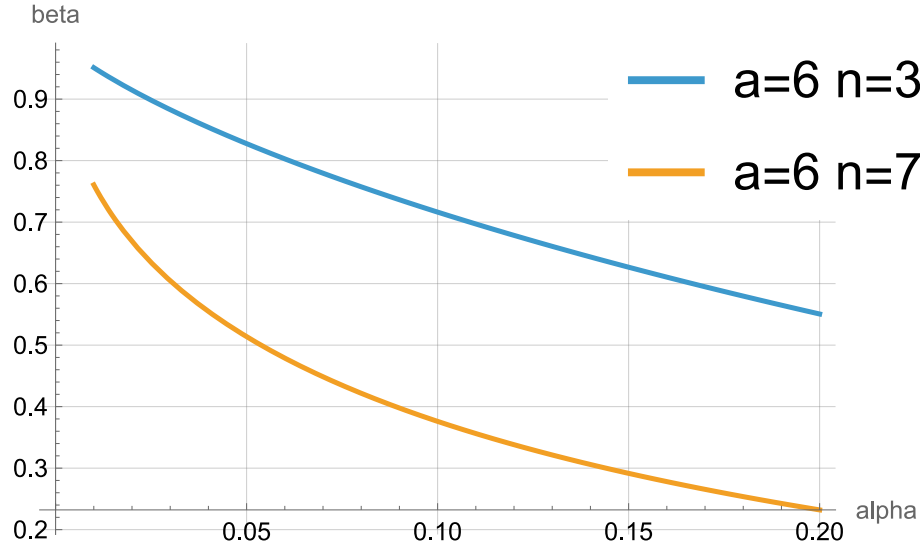
$$F_{\alpha}(df_{FAC}, df_E)$$

$$\boxed{d / \sigma}$$

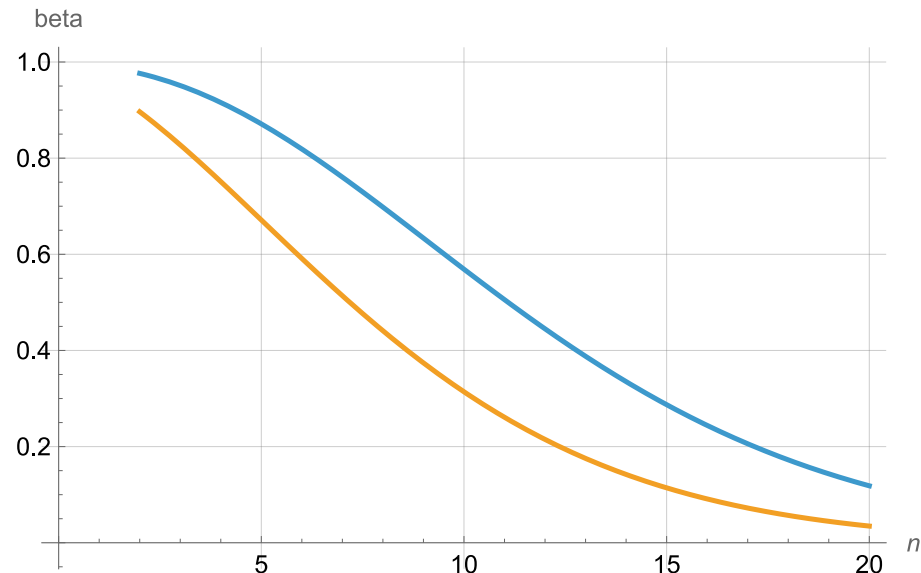
Power



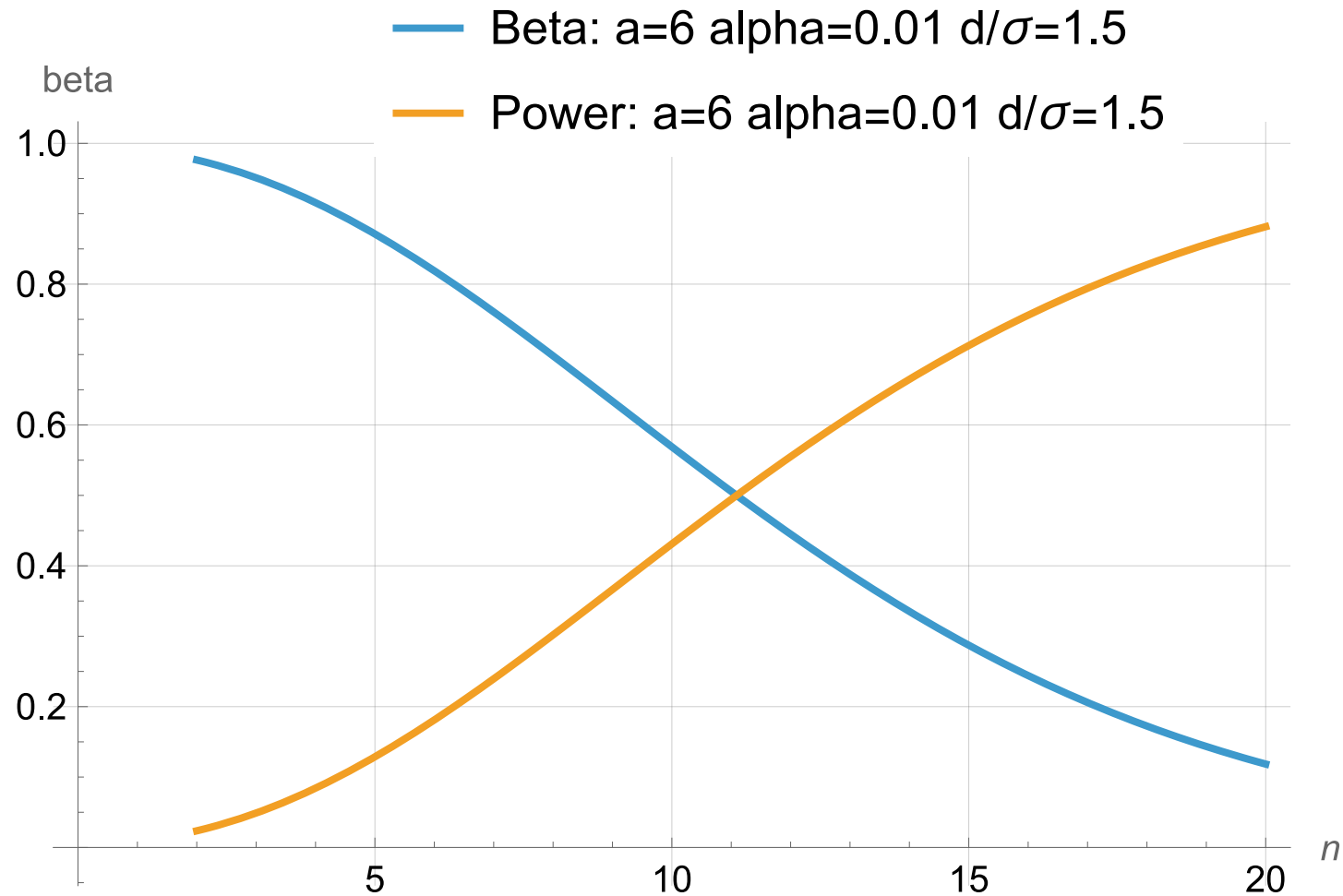
Beta depends on alpha and n



— $a=6$ $\alpha=0.01$ $d/\sigma=1.5$
— $a=6$ $\alpha=0.05$ $d/\sigma=1.5$

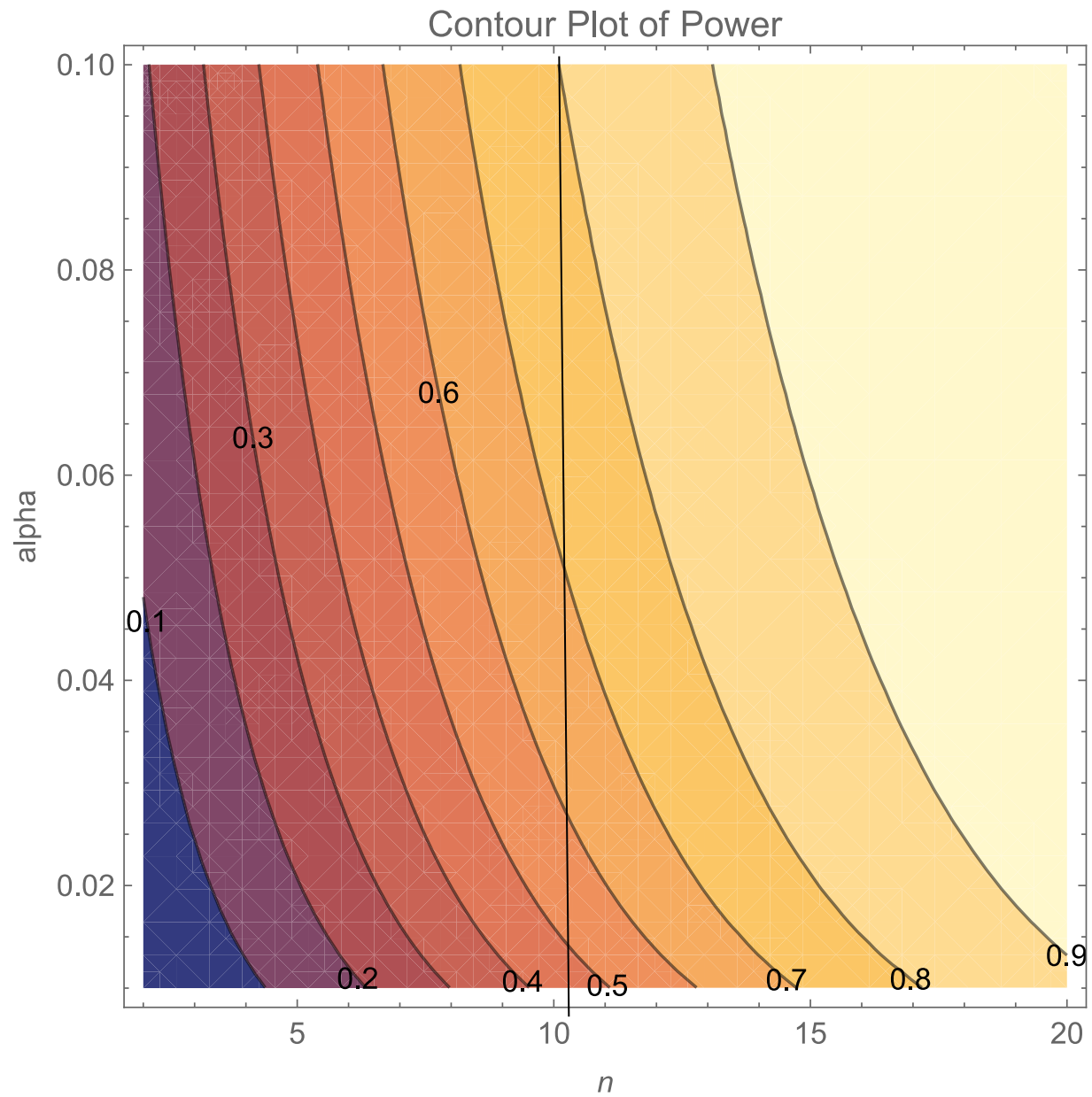


Power=1-beta



Power

$a=6$
 $d/\sigma=1.5$



C) Maximum Standard Deviation Increase G

Let's suppose $\tau_i \neq 0$ for some i

$$Y_{ij} = \mu + \tau_i + \varepsilon$$

$$E(Y_{ij}) = E(\mu) + E(\tau_i) + E(\varepsilon) = \mu$$

$$\begin{aligned}\sigma_Y^2 &= E\left[\left(Y_{ij} - E(Y_{ij})\right)^2\right] = E\left[\left(\tau_i + \varepsilon\right)^2\right] = \\ &= E\left[\tau_i^2 + 2\tau_i\varepsilon + \varepsilon^2\right] = \frac{\sum_i \tau_i^2}{a} + \sigma^2\end{aligned}$$

Ignoring the **Factor Effects** implies that we accept a dispersion higher than the real/potential one.

We specify the **Standard Deviation Increase G**

$$G^2 = \frac{\sigma_Y^2}{\sigma^2}$$

$$\left(\frac{PCR}{PCR_Y} \right)^2 = \left(\frac{\sigma_Y}{\sigma} \right)^2 = G^2$$

$$G^2 = \frac{\sum_i \tau_i^2}{a\sigma^2} + 1 \quad \rightarrow \quad \sum_i \tau_i^2 = \sigma^2 a (G^2 - 1)$$

$$\delta_{FAC} = n \frac{\sum_i \tau_i^2}{\sigma^2} = an (G^2 - 1)$$

You can use the sw based on the minimum difference using:

$$\frac{1}{2} \left(\frac{d}{\sigma} \right)^2 = a (G^2 - 1) \quad \rightarrow \quad \frac{d}{\sigma} = \sqrt{2a (G^2 - 1)}$$

JMP v17 Approach

JMP sw is based on the **Minimum Difference of Interest** approach.

Let us consider the previous example

$$a = 4 \quad n = 3 \quad \delta_{FAC} = 36 \quad \alpha = 0.05$$

$$\delta_{FAC} = \frac{n}{\sigma^2} \sum \tau_i^2 = \frac{n}{2} \left(\frac{d}{\sigma} \right)^2 \quad \frac{d}{\sigma} = \sqrt{2 \frac{\delta_{FAC}}{n}} = \sqrt{2 \frac{36}{3}} = 4.89898$$

Pay attention:
too many decimals for practical purposes

$$\beta = 1 - 0.981583 = \mathbf{0.01847}$$

Command: DOE->Sample Size Explorer->Power-> Power for ANOVA

Power Explorer for ANOVA

Explorer Settings

Test Specification

- ☐ Group means
- ☐ Between-group variance
- ☒ Maximum Difference
 - ☒ Worst Case
 - ☐ All but One

Preliminary Information

Alpha

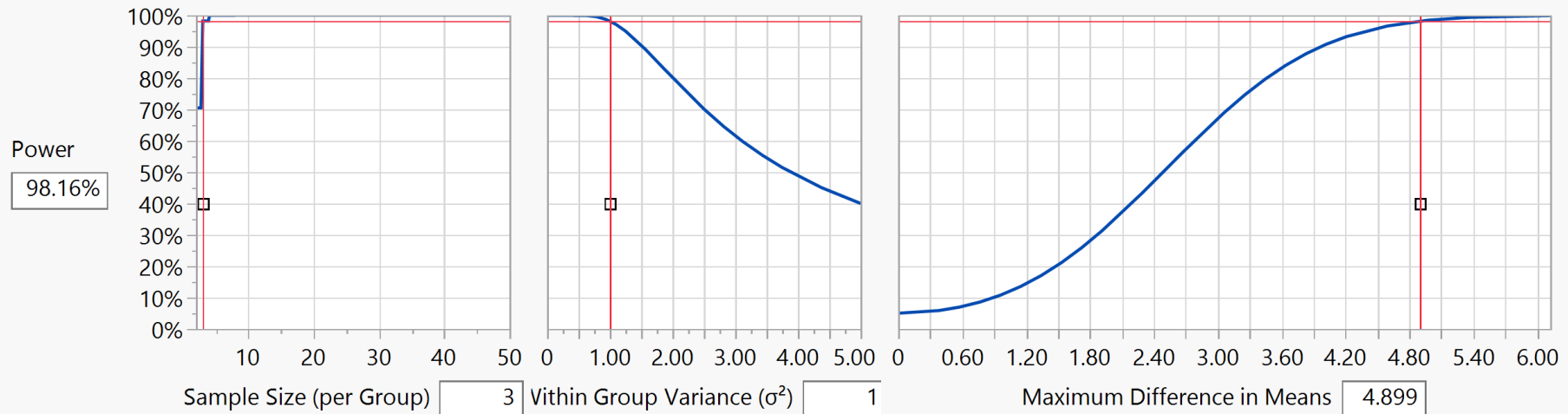
Number of groups

$$a = 4 \quad n = 3 \quad \delta_{FAC} = 36 \quad \alpha = 0.05$$

$$\frac{d}{\sigma} = 4.89898$$

Profiler

Solve for:



Save Settings

Help

Saved Settings

$$\beta = 1 - 0.981583 = \mathbf{0.01847}$$

Problem: how to find optimal α , β and n ?

- Let us assume we know:
 - a (number of factor levels)
 - d/σ (Violation of interest)
 - c_1 loss if we reject H_0 when H_0 is true
 - c_2 loss if we fail to reject H_0 when H_1 is true
 - c_3 cost of each run
- Find α , β and n (number of replicates for each level) that minimize the **Expected Loss** (Objective function)
- We could add two constraints: $\alpha \leq \alpha_0$, $\beta \leq \beta_0$

Bechhofer Method

(Correct Identification)

- Specify n to be, **reasonably**, sure that the **maximum observed sample mean** belongs to the **population with the true maximum mean**

$$\mu_{[a]} \rightarrow \bar{y}_{[a]}.$$

- Hypotheses:** normal, independent and constant variance

$$\lambda = \mu_{[a]} - \mu_{[a-1]}$$

$$1 - \gamma = \Pr \left\{ \mu_{[a]} \rightarrow \bar{y}_{[a]} \right\}$$

a	$1-\gamma = 0.90$	0.95	0.99
2	1.81	2.33	3.29
3	2.23	2.71	3.62
4	2.45	2.92	3.80
5	2.60	3.06	3.92
6	2.71	3.16	4.01
7	2.80	3.24	4.09
8	2.87	3.31	4.15
9	2.93	3.37	4.20
10	2.98	3.42	4.25

- Example**

$$a = 4 \quad 1 - \gamma = 0.90$$

$$\sigma = 2 \quad \lambda = 1$$

$$2.45 = \frac{\lambda}{\sigma} \sqrt{n}$$

$$n = \left(2.45 \frac{\sigma}{\lambda} \right)^2 \simeq 24$$

Effects of hypothesis violation

- Not Normal Data (but well shaped)
 - F Test is robust
 - Scheffè is robust if we are interested to all **pair** comparisons
- Non-constant variance
 - F Test is robust if n_i are constant
- Residual Autocorrelation
 - Serious consequences.

Nonparametric ANOVA

- When data are not normal (or they have no-constant variance) after transformations
- We test the equality of **Treatment Medians** (shift location)
- Hypotheses:
 - Independent data
 - Continuous distributions with the same general form (no bimodal)
- **Kruskall -Wallis Test** (generalization of the Mann-Whitney Test also called Two Sample Wilcoxon Rank Sum Test)
- We could use several alternative tests (i.e. the MOOD's Median Test which is robust to the outliers but less powerful than Kruskal-Wallis test)

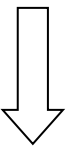
Experimental data

Replicates

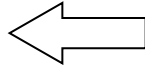
Factor Level	1	2	3	4
1	5.0	7.8	4.1	3.1
2	10.0	11.1	9.9	8.5
3	4.2	3.3	5.4	3.0

We use the observation
Rank Transformation

In this example we don't
have **ties**



Level	1	2	3	4
1	6	8	4	2
2	11	12	10	9
3	5	3	7	1



Compute the Rank means

Factor	1	2	3	4	$\bar{R}_{i.}$
1	6	8	4	2	5
2	11	12	10	9	10.5
3	5	3	7	1	4

$\bar{R}_{..} = 6.5 = \frac{N+1}{2}$

If the row means are different from the general mean,
probably the treatments have different location

If the treatment median are equal

$$H_0 = \frac{12}{N(N+1)} \sum_i n_i (\bar{R}_{i.} - \bar{R}_{..})^2 \sim \chi^2(a-1)$$

Note: $n_i \geq 5$

In our example

$$H_0 = 7.53$$

$$\chi_{0,05}^2(2) = 5.99$$

Then $\rightarrow$ **Reject**

JMP analysis

Fit Y by X->Nonparametrics->Kruskal-Wallis

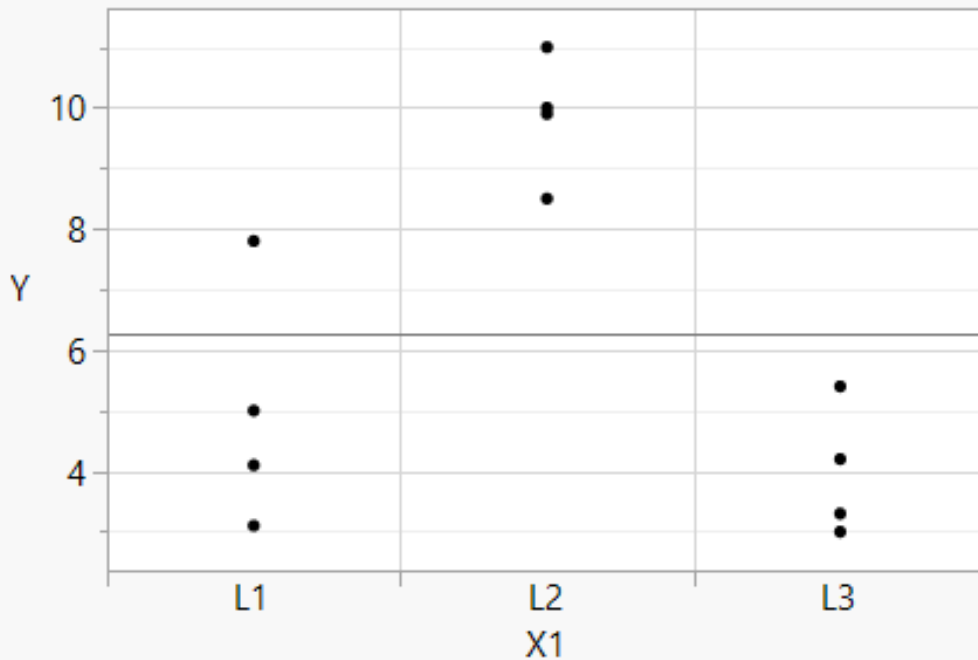
Wilcoxon / Kruskal-Wallis Tests (Rank Sums)

Level	Count	Score Sum	Expected Score	Score Mean	(Mean-Mean0)/Std0
L1	4	20.000	26.000	5.0000	-0.934
L2	4	42.000	26.000	10.5000	2.633
L3	4	16.000	26.000	4.0000	-1.613

Kruskal-Wallis Test, ChiSquare Approximation

ChiSquare	DF	Prob>ChiSq
7.5385	2	0.0231*

Small sample sizes. Refer to statistical tables for tests, rather than large-sample approximations.



- **Ties**: We use the **Rank averages** (limited number of ties)
- If we have a lot of ties **Lehman** suggests the following correction (Lehman “Nonparametrics: statistical methods based on ranks”, Holden Day)

$$H_0^* = H_0 \frac{1}{1 - \sum_{l=1, VD} \left(\frac{d_l^3 - d_l}{N^3 - N} \right)}$$

$VD \equiv$ Number of distinct values

$d_l \equiv$ Number of data at the l -th distinct value

- Note 1 for $n < 5$ specific tables exist
- Note 2 for $a=2$ the test is equal to the **Mann-Whitney Test**
- Note 3 the test is equivalent to a standard F Test on Ranks